

5

Finite Active Arrays

5.1 INTRODUCTION

In the previous chapter we studied surface waves on passive structures as for example finite FSSs. They were excited by an incident plane wave. We observed that a finite FSS in addition to the Floquet currents excited directly by the incident plane wave could also support surface waves. These would radiate and thereby lead to an increase in the scattered field; that is, the RCS could be larger than expected. The scattered field associated with the surface waves could be significantly reduced by resistively loading one or more columns at the edges of the finite FSS. This approach would leave the Floquet currents in the rest of the FSS unaffected; that is, the transmission and reflection properties of the FSS were basically left intact.

In this chapter we shall investigate the effect of surface waves on active periodic structures—like, for example, finite arrays. The most significant difference between the two cases is that while the first is excited by an incident plane wave, the second either is excited by a generator or is delivering the incident energy to a conjugate-matched amplifier, not just a reactive load. The important point here is that the generators as well as the amplifiers have impedances with a significant resistive component connected to the terminals of each element. This resistive component will as shown in Fig. 1.5a exhibit a significant attenuation on any potential surface wave. In fact, the amplitude of the surface waves on an active array is typically so low that the radiation from this component is of no great concern. However, as discussed in Section 1.4, even a small

amount of surface waves leads to some variation of the scan impedance of the individual elements. Thus, since it is desirable to use the same matching network for each element, precise matching can be a problem. This mismatch is typically so small that it does not adversely affect the received power. However, it may present a problem, if we are concerned with the RCS of the array as discussed in Chapter 2.

Another difference between the passive and active cases is worth mentioning. The first typically is comprised of FSS arrays with either slot and/or wire elements. They may be located in a stratified medium, but they will in general not directly contain a groundplane. In contrast, the active array will usually consist of a single array of either the wire or slot type, and they may also be located in a stratified medium but they are almost always provided with a groundplane.

The groundplane serves essentially two purposes. First of all it ensures that we have only a single mainbeam, not two. Second, as discussed in Chapter 2, the groundplane can lead to a significant reduction of the RCS of an active array. However, as discussed in Section 2.9, the area of the groundplane relative to the area of the active dipoles is crucial from an RCS point of view. Thus, the exact modeling of the finite groundplane becomes important. We shall discuss this issue in the next section.

5.2 MODELING OF A FINITE $\times$ INFINITE GROUNDPLANE

Our finite $\times$ infinite groundplane is comprised of an FSS with an area approximately matching the area of the active elements. It resonates at the center frequency, yielding a reflection coefficient equal to -1 like a perfect groundplane; however, some leakage will take place at other frequencies. Thus, it should be designed to have as broad a bandwidth as possible, or, alternatively, it may be retuned to other frequencies as deemed necessary.

Examples of FSS groundplanes are shown in Fig. 5.1. More specifically, we show a typical practical version in Fig. 5.1a. Each column consists of closely spaced straight wire sections interlaced as shown. Associated with the central part of each wire section is some inductance and associated with the small gap between them is some capacitance. Thus, the equivalent circuit will look approximately as shown in Fig. 5.1b. At resonance the reactance of such a sheet is zero; that is, it will act as a groundplane as noted already above.

Alternatively, the practical layout may also be executed by using actual lumped elements as shown in Fig. 5.1c. In fact, the inductances can consist of straight wire sections as shown in Fig. 5.1d. Similarly the capacitors may be partly lumped. They are simply in parallel with the end capacitors between adjacent element tips. This capacitance can be much more significant than many people expect (for further comments concerning this subject see references 89 and 90). Thus, the lumped capacitors act more like a fine-tuning device. See also Section 6.4.

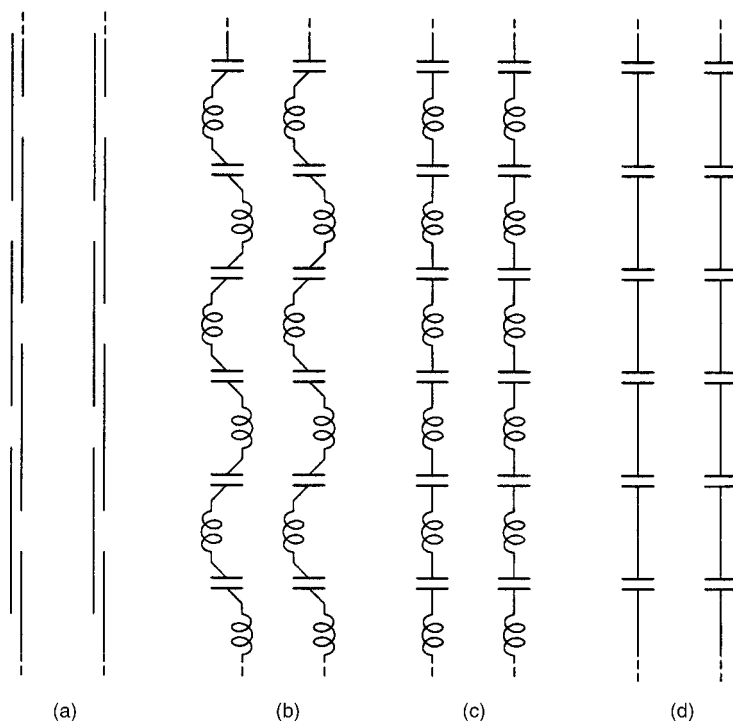


Fig. 5.1 Modeling of the reflecting groundplane by using FSS's of various forms.

5.3 FINITE $\times$ INFINITE ARRAY WITH AN FSS GROUNDPLANE

In Chapter 2 we discussed the backscatter from large arrays in general without considering the possible effect of surface waves or edge scattering. We shall investigate both of these phenomena in this chapter. As a simple but very instructive introduction to the latter, let us now consider the bistatic scattering pattern for a finite $\times$ infinite array of active dipoles backed by a finite $\times$ infinite FSS “groundplane” of the type discussed in Section 5.2. It illustrates many of the ideas and concepts discussed in Chapter 2. The calculated curves were obtained from the SPLAT program as were the curves in Chapter 4.

A top view of the array layout with the FSS groundplane is shown in the insert of both Figs. 5.2 and 5.3. It is comprised of 20 columns of active elements and twice as many FSS elements. The reason for more passive than active elements is simply to make the FSS groundplane more broadband. (In general, the closer the elements, the greater the bandwidth; cf. the “Gangbuster” array in reference 91.)

Putting the FSS elements closer together raises the resonance frequency somewhat. Since the elements cannot be made longer without disturbing the periodicity, all the passive elements have instead been loaded with $Z_L = j208$ ohms (*note*: Here we do not use interlaced elements of the type shown in Fig. 5.1a, just

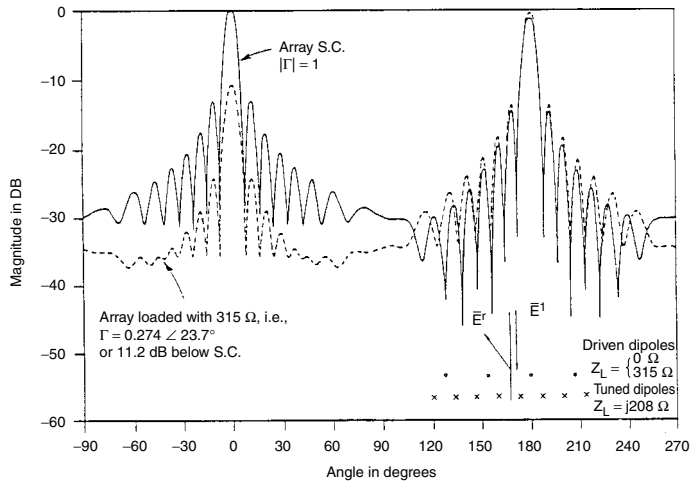


Fig. 5.2 Calculated bistatic scattered field obtained from the SPLAT program of a finite array of dipoles backed by a finite FSS "groundplane" for an incident plane wave arriving at broadside (0°). Two curves: One for the driven dipoles short-circuited (S.C.) and another when loaded with $Z_L = 315$ ohms.

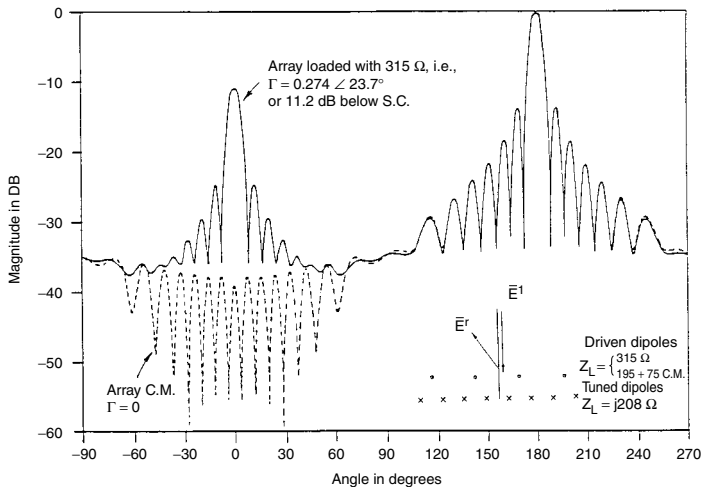


Fig. 5.3 Calculated bistatic scattered field of a finite array of dipoles backed by a finite FSS "groundplane" for an incident plane wave arriving from 0° . Two curves: One when loaded with $Z_L = 315$ ohms (like Fig. 5.2) and another when conjugate-matched. Obtained from the SPLAT program.

single straight elements). Actually this kind of tuning does not enhance the bandwidth, but we are here at present only concerned with one frequency (obviously these curves were obtained early on before we had reached our present level of sophistication).

A plane wave is incident broadside to this array—that is, at 0° . The bistatic scattered field is obtained from the SPLAT program in the entire range from -90° to 270° . Furthermore, we show the bistatic fields for various load conditions of the active elements.

First in Fig. 5.2 we show two curves, one where the active elements are short-circuited and denoted S.C. and another where $Z_L = 315$ ohms. Note that for $Z_L = 0$ the backscattered field looks like it is coming from a groundplane the size of the array. Furthermore, for $Z_L = 315$ ohms and with $Z_A = 195 - j75$ ohms we obtain a reflection coefficient

$$\Gamma = \frac{Z_L - Z_A}{Z_L + Z_A} = \frac{315 - 195 + j75}{315 + 195 - j75} = 0.274 \angle 23.7^\circ,$$

or 11.2 dB below the short-circuited load as actually observed in Fig. 5.2.

Furthermore, in Fig. 5.3 we show the bistatic scattered field when $Z_L = 195 + j75$ ohms—that is, conjugate-matched. We also repeat the case $Z_L = 315$ ohms to facilitate comparisons.

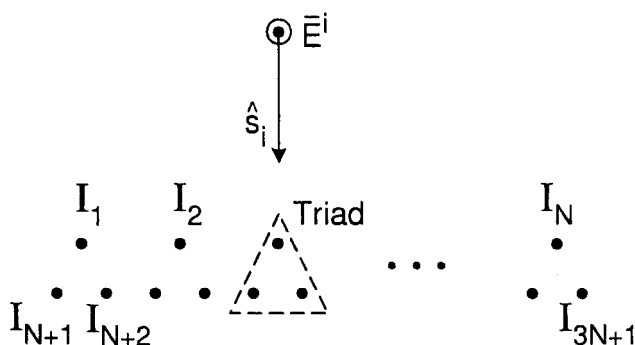
The conjugate-matched case is particularly interesting. According to Section 2.9 and more precisely Fig. 2.12, Example II, we should expect the backscattered field to be approximately equal to zero. Inspection of Fig. 5.3 reveals a level of ~ 38 dB below a flat plate and not ∞ dB. In the next section we will show that this discrepancy is caused by an edge effect. We shall in fact devise an approach that not only very precisely pinpoints where the excess scattering is coming from but also how to alleviate it. Finally we further note in Figs. 5.2 and 5.3 that the forward-scattered field at $\pm 180^\circ$ is practically independent of the load impedance Z_L of the active elements as was predicted in Section 2.9.

This is also an excellent place to remind the reader that the *total* field in the forward direction is the sum of the incident field and the forward-scattered field. The former is merely a plane wave with an amplitude that is conceptually independent of its distance from the array in the forward direction. However, the latter will, as all fields scattered by an object of finite extent, be attenuated as we move away from the antenna. Thus, although the two components cancel each other approximately right behind the groundplane and produce “darkness” as one would expect, the incident field will soon dominate the scattered field. The result is that the “shadow” from the antenna will soon be blurred and eventually practically disappear. In other words, beating the stealth concept by observing the forward-scattered field might not be as easy as some individuals like to think.

5.4 MICROMANAGEMENT OF THE BACKSCATTERED FIELD

We are now going to show how we can pinpoint exactly any scattering anomalies in an array and how to alleviate it. To illustrate our approach, we are again going to consider a finite array of active dipoles backed by a finite FSS groundplane as shown in Fig. 5.4, top. It is being exposed to an incident plane wave at broadside.

Modeling and Approach Two-Dimensional



The finite groundplane is modeled by a
finite reflecting FSS (see Fig. 5.1)

Approach

1. Calculate all element currents including all mutual impedances (rigorous and exact).
2. Calculate all fields from each triad (see above figure) and plot in the complex plane; see Fig. 5.6.

Fig. 5.4 Modeling of a finite array with a finite groundplane.

We start by calculating all column currents in the *entire* array configuration. Note that we take into account all mutual impedances between all wire segments; that is, the calculated currents are exact within the accuracy of the Method of Moments.

Most readers would at this point be tempted to obtain the entire scattered field by simply adding the fields reradiated by each wire section. Although this would be technically correct, it is precisely what not to do. Rather, the scattered fields should be grouped together in a meaningful way.

More specifically, for the case shown in Fig. 5.4 we add the fields from each active column to the fields from the two passive columns located right behind it

Radiation and Scattering Pattern for a Typical Triad

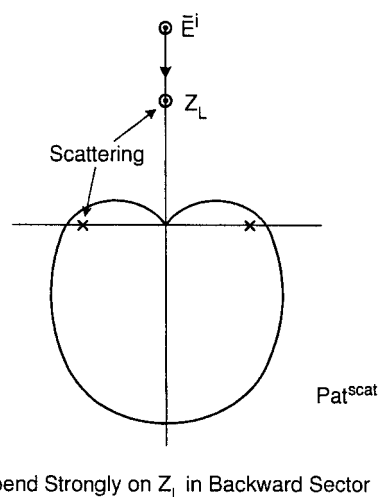
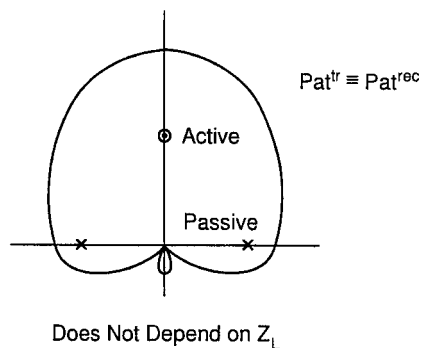
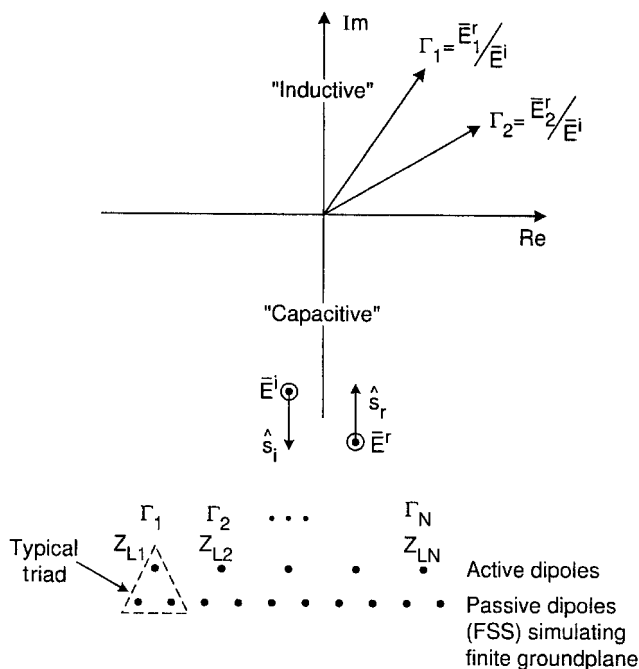


Fig. 5.5 Top: Typical transmitting or receiving pattern for a typical triad comprised of an active column of dipoles and two columns of passive FSS elements. Bottom: Typical scattering pattern from the same triad as above.

(we call such a configuration a triad). The reason for combining the fields in that way is illustrated in Fig. 5.5. At the top we show a typical radiation pattern for a single active column backed by two passive columns. This radiation pattern is similar to a cardioid pattern with the main beam pointing upward. At the bottom we show the scattered pattern for the same configuration when exposed to a plane wave incident from above. Note that the main beam is pointing downward. It is, however, not merely equal to the transmitting pattern rotated 180° . In particular we note that the transmitting pattern is completely independent of the load impedance Z_L while the scattering pattern in the backward sector is highly dependent on Z_L (see also the discussion in Section 2.9 as well as Problem 2.6).

Analyzing the Backscatter from a Finite Array



The individual loads Z_{L_n} can be adjusted to yield
 $\Gamma_n \sim 0$ individually; see Fig. 5.8.

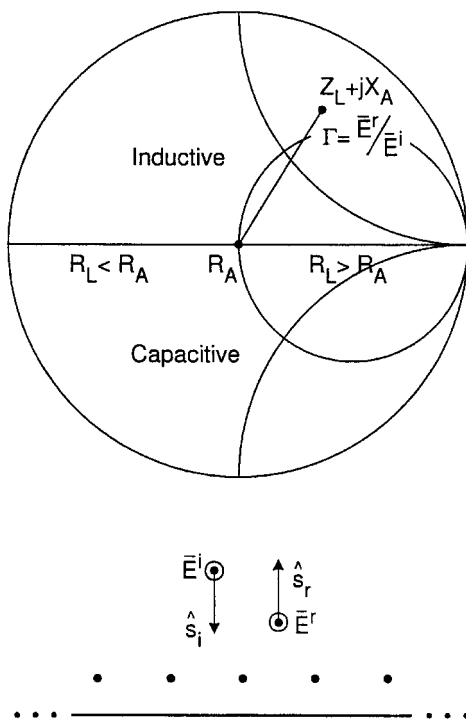
Fig. 5.6 A plane wave incident upon a finite array of triads produced the backscattered signals E_n^r . By plotting $\Gamma_n = E_n^r / E_i$ in the complex plane, each triad can be analyzed and adjusted separately.

Let us denote the field from each triad by $\bar{E}_n^r$. We then define the reflection coefficient for each triad as $\Gamma_n = E_n^r / E_i$, where E_i denotes the incident field. The “trick” is now to plot Γ_n in the complex plane as shown, for example, in Fig. 5.6. Note that the Γ_n ’s are different for each triad because Floquet’s Theorem is no longer valid for a finite array.

Our next step is inspired by the Smith chart. (Some readers are happy to see this ingenious device “back in” in our curriculum. It was never “out” as far as I am concerned. See Chapter 6 and Appendices A and B and you will understand my devotion.).

Consider an infinite array of active dipoles loaded with Z_L and backed by an infinite FSS groundplane as shown in Fig. 5.7, bottom. Let us plot the reflection coefficient $\Gamma = \bar{E}^r / \bar{E}^i$ for the infinite array in a complex plane as, for example,

Backscatter from an Infinite Array



By adjusting Z_L we can obtain $\Gamma \sim 0$, i.e.,
 $\sigma_{\text{tot}} \sim 0$.

Advantage of using Smith chart: It tells us
 HOW to adjust Z_L to obtain $\sigma_{\text{tot}} \sim 0$.

Fig. 5.7 A plane wave incident upon an infinite array of active elements in front of an infinite groundplane. By plotting the reflected field in a complex plane (in this case a Smith chart) we can adjust the load impedances Z_L of the active dipoles such that the reflected field disappears in the backscatter direction.

a Smith chart normalized to R_A as shown in Fig. 5.7, top. The complex reflection coefficient Γ is then determined by the location of $Z_L + jX_A$ as shown in Figs. 2.3 and 2.6. (In fact, if $jX_A = 0$, the reflection coefficient Γ reduces to the ordinary reflection coefficient as explained in Fig. 2.3.)

The beauty of the Smith chart (one of many) is now that the location of $Z_L + jX_A$ will tell us how to adjust Z_L to obtain $\Gamma = 0$. For example, if $Z_L + jX_A$ is located in the upper right half of the Smith chart as shown in Fig. 5.7, we know

immediately that the inductance jX_L should be reduced and so should R_L in order to minimize $|\Gamma|$. The idea is now that we can at least in principle use a similar technique to reduce the individual reflection coefficients $|\Gamma_n|$ shown in Fig. 5.6.

However, while the adjustment of Z_L can be done precisely in the Smith chart, this is not quite the case for $|\Gamma_n|$ shown in Fig. 5.6. One reason being that the field reflected from the infinite array is a simple plane wave (we assume the evanescent waves have died out), while the fields from the triads are a combination of Hankel functions. Should we call this new chart a Hankel chart? Somebody could work it out and cover himself with fame and glory. See also comments in Section 5.8.

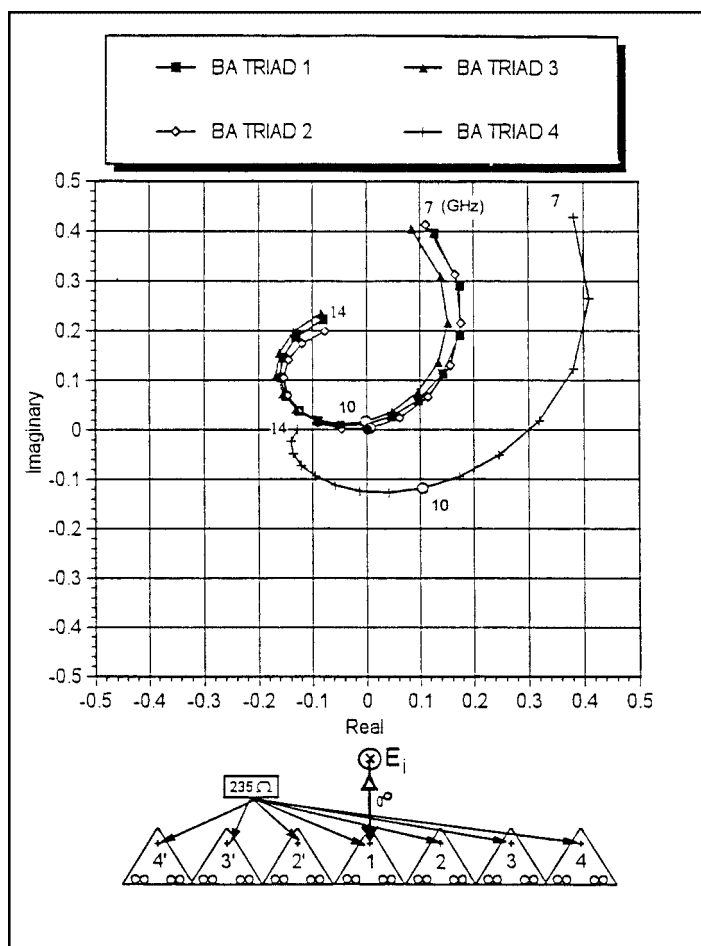


Fig. 5.8 The backscattered fields from a finite array of 7 triads plotted in the rectangular complex plane. All active dipoles are loaded with the same impedance equal to 235 ohms. This value leads to nearly no backscatter for all the triads at $f = 10.0$ GHz except for the edge columns 4 and 4'. By adjusting the loads for these two separately, the backscatter from these could also equal zero. (From Johnson [75].)

A typical plot of the fields from 7 triads is shown in Fig. 5.8. We immediately note the two most important problems in a finite array:

1. The fields scattered from each of the two outer columns (denoted 4 and 4') are different and for the load impedance $Z_L = 235$ ohms larger than all the other triads at 10 GHz. However, if we change the load impedances for each of the outer columns by adding a little inductance and by lowering the real part, we can reduce the edge scattering that is so prominent in Fig. 5.3 for conjugate match. (Inspection of the scattering pattern in the figure with fine lobes of almost equal amplitudes is a strong indication that the scattering does originate on the edges.)
2. The second problem is somewhat more intriguing, namely the fact that the scattering from the various triads varies erratically ("jitters").

In principle we could match the individual columns with the proper but different load impedances and obtain zero total scattering. However, this approach is deemed too cumbersome and impractical.

We would prefer to solve this problem in a more elegant and universal way. We suspect that the impedance variation from column to column is related to surface waves (or at least a degenerate form of a surface wave). In the next sections we shall identify the problem and show how to reduce their effect.

5.5 THE MODEL FOR STUDYING SURFACE WAVES

We shall next study surface waves on active arrays with a finite FSS ground-plane. Our model will be similar to the one used in the previous section—except that in order to properly study surface waves, the model must be considerably wider.

The fundamental problem is now that all finite periodic structures may exhibit strong presence of surface waves at least at some frequencies as discussed in Chapter 4. We may envision that the finite FSS groundplane alone shows surface waves in one frequency band and the active array possibly in another. However, when the active array is placed adjacent to the FSS groundplane, we would expect both of these frequency bands to change and, possibly, to degenerate into a single frequency band. From a practical point of view, it is of course the surface waves on the combined structure that are most important.

Although it is feasible that an actual practical array could be built as modeled here namely by using a finite FSS groundplane, it is more likely that the ground-plane will consist of just a finite piece of perfectly conducting sheet metal with holes so small that no surface wave can exist. In that case the possible surface waves will depend only on the active array elements and is therefore less complex. More specifically, if we can learn to control the surface wave phenomena on arrays with finite FSS groundplanes, there is good reason to expect that we can do the same when using a finite perfectly conducting groundplane.

Calculating the effect of a finite conducting groundplane may not be as simple as some individuals think. Merely using the scattering mechanisms of a semi-infinite groundplane may be jeopardized by the presence of the active array (simply “trading” these coefficients as baseball cards should not be encouraged). Also, use of plates as is ordinarily done in the Method of Moments often leads to leaks behind the groundplane which are avoided completely at the resonant frequency of the FSS groundplane.

Thus, in the following we shall use the finite FSS groundplane and accept the greater complexity due to the more intricate presence of surface waves.

In the next section we shall examine these surface waves in more detail and also ways to alleviate them.

5.6 CONTROLLING SURFACE WAVES ON FINITE FSS GROUNDPLANES

In this section we shall investigate surface waves on finite FSS groundplanes without any active dipoles. Typically they are comprised of interlaced wire elements each of length 1.35 cm as shown schematically in the insert of Fig. 5.9, bottom. The frequency scattering diagram is also shown in Fig. 5.9, top, for angle of incidence equal to 45° . To the left we show the frequency range 2.0–6.7 GHz, and to the right we show the range 6.8–12.0 GHz. It is obvious by inspection that surface waves are significant in the lower frequency range but not in the upper range, in accordance with our observations in Chapter 4, Section 4.9.3.

By inspection of Fig. 5.9 we chose the frequency 5.7 GHz (more or less arbitrarily) as a representative of a range of frequencies with strong surface waves. The actual column currents obtained from the SPLAT program are shown in Fig. 5.10. Indeed, we observe very strong surface waves at that frequency.

We shall attempt to suppress these surface currents similarly as we did in Chapter 4, namely by loading the edge columns only with resistors located at the center of each element. Two examples are shown in Figs. 5.11 and 5.12, respectively. In the first case we have loaded the three outermost columns with the load resistors 200, 100, and 50 ohms, respectively, as shown in the insert of Fig. 5.11, bottom. Similarly, the example in Fig. 5.12 has the seven outermost columns loaded with resistors as indicated in the figure. In the following we shall refer to these as the lightly and heavily loaded cases, respectively.

We observe a significant reduction of the surface waves as compared to the “wild” case shown in Fig. 5.10, in particular for the heavily loaded case shown in Fig. 5.12. Neither of these two cases represents in any way the optimum. Undoubtedly, further improvement is possible. However, we should not lose sight of the fact that our “final” product in reality is the finite FSS groundplane including the active loaded columns in front. Since such a combination can be expected to have strong surface waves in a frequency range somewhat different from what we saw in Figs. 5.9 and 5.10, we shall postpone further investigation of this subject until the next section.

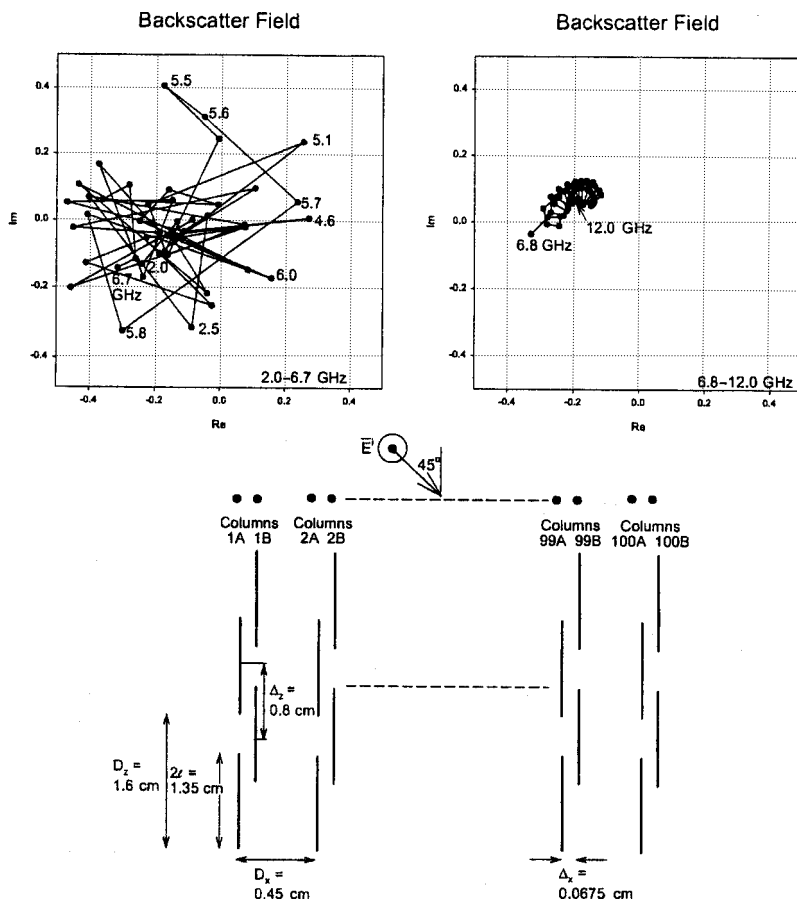


Fig. 5.9 The field reflected from the finite FSS array shown in the middle and plotted in the complex plane as a function of frequency (scattering diagram). Top left: The frequency range 2.0–6.7 GHz where surface waves are prevalent. Top right: The frequency range 6.8–12.0 GHz shows no sign of strong surface waves.

Finally we should remind ourselves that a practical antenna probably will be made with a groundplane comprised of a finite perfectly conducting groundplane rather than using a finite FSS (except perhaps at the edges; see later). Thus, in that event our interest in surface waves associated with the finite FSS groundplane has been reduced to be of merely academic interest.

5.7 CONTROLLING SURFACE WAVES ON FINITE ARRAYS OF ACTIVE ELEMENTS WITH FSS GROUNDPLANE

In the previous section we found surface waves on the FSS groundplane most prevalent below 6.7 GHz. Let us now examine the more complicated problem

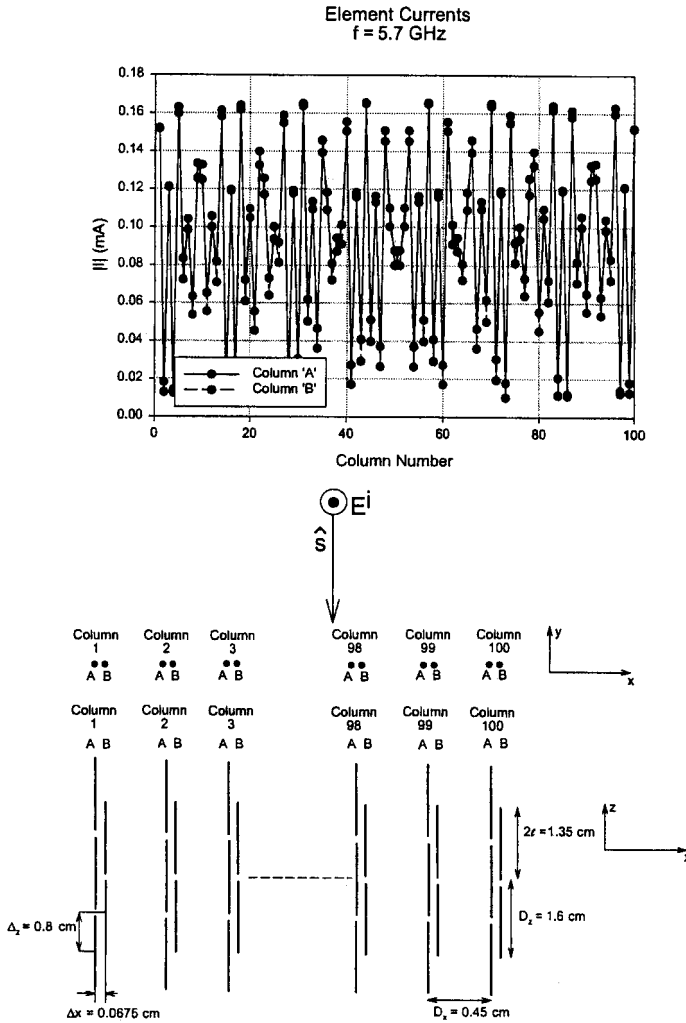


Fig. 5.10 The element currents (column) for the finite FSS array without loading shown at the bottom. Incident field at broadside. Frequency 5.7 GHz where strong surface waves are present.

where our structure consists of the same finite FSS groundplane as above but with active elements placed in front of it.

This structure will be tested for surface waves at three frequencies: the same low frequency as before, $f_L = 5.7 \text{ GHz}$; a frequency in the middle, $f_M = 7.8 \text{ GHz}$; and the highest frequency, $f_H = 10 \text{ GHz}$.

5.7.1 Low Test Frequency $f_L = 5.7 \text{ GHz}$

We show in Fig. 5.13 the column currents on the lightly loaded FSS groundplane shown earlier in Fig. 5.11. The active elements are loaded with $R_L = 100 \text{ ohms}$,

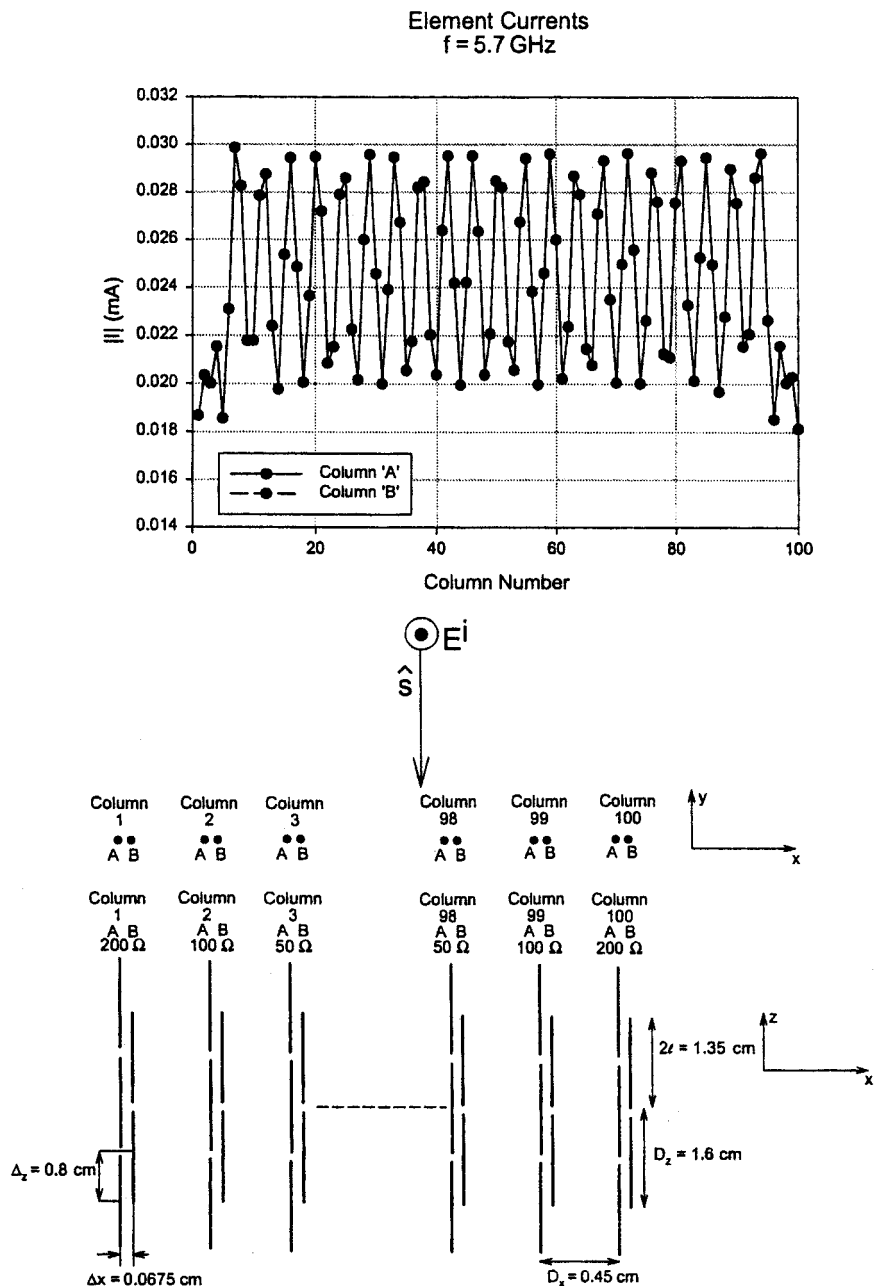


Fig. 5.11 The element currents for the same finite FSS array as shown in Fig. 5.10, but with light loading as shown at the bottom. Note the dramatic reduction of current oscillations—that is, weaker surface waves. Also note the change of scale compared to Fig. 5.10.

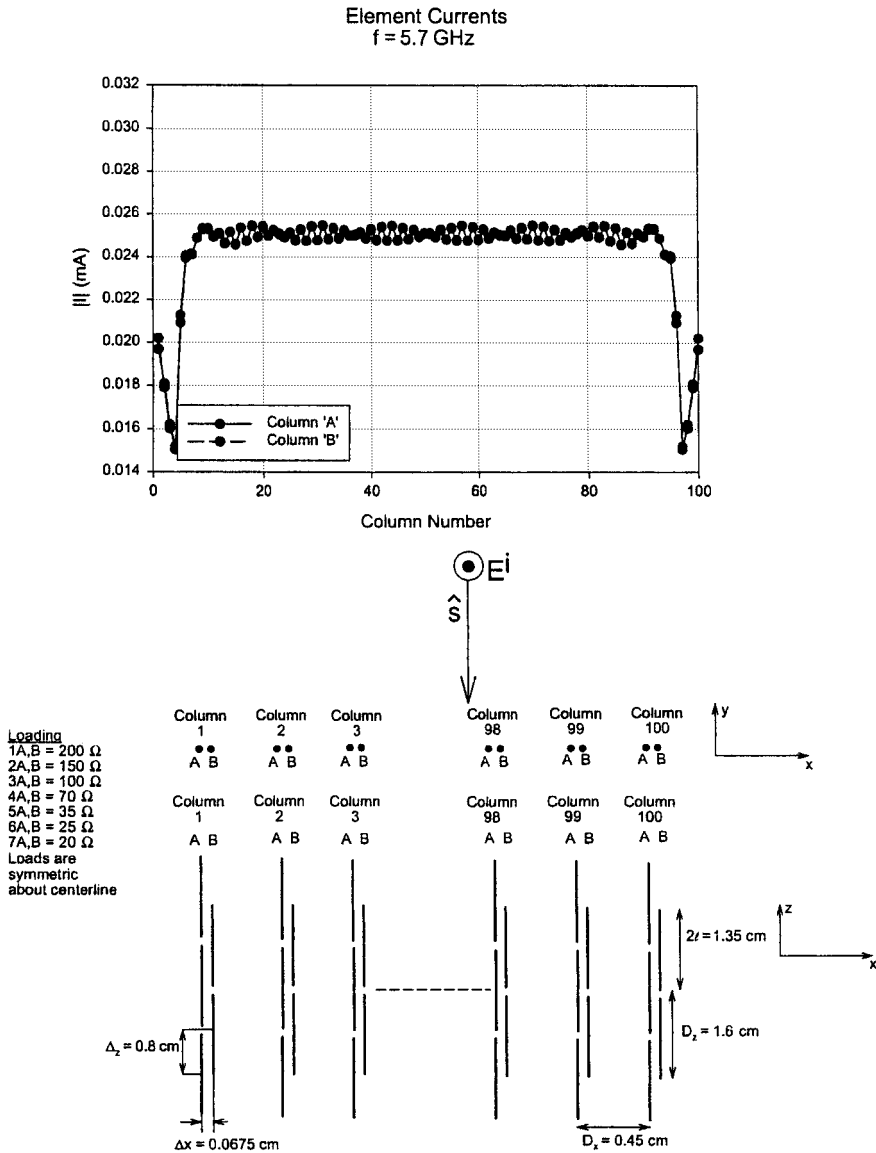


Fig. 5.12 The element currents for the same finite FSS array as shown in Figs. 5.10 and 5.11, but heavily loaded as shown at the bottom. Note the further reduction of the surface waves.

which is actually rather low considering that the antenna resistance R_A is ~ 235 ohms (see later). Thus, we also show in Fig. 5.14 the lightly loaded FSS groundplane case but where the active elements are now loaded with $R_L = 200$ ohms. We note that the current variations from element to element are of approximately the same magnitude in both cases and although small, it

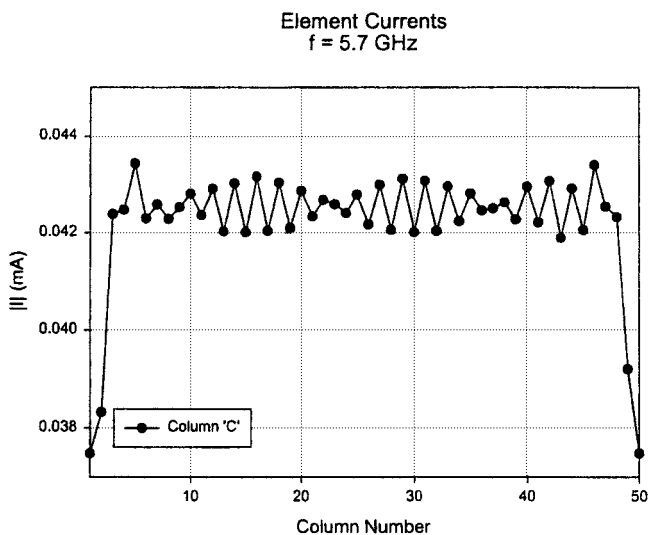


Fig. 5.13 The element currents of the active dipoles placed in front of the lightly loaded FSS array shown in Fig. 5.11. The active dipoles are all loaded with 100 ohms. Frequency is 5.7 GHz.

would be desirable to have them even lower. Note also that the average current is reduced from ~ 0.043 to ~ 0.032 as expected (see Section 1.4).

Thus, we show in Fig. 5.15 the heavily loaded FSS groundplane case and where the active elements again are loaded with $R_L = 200$ ohms. We observe

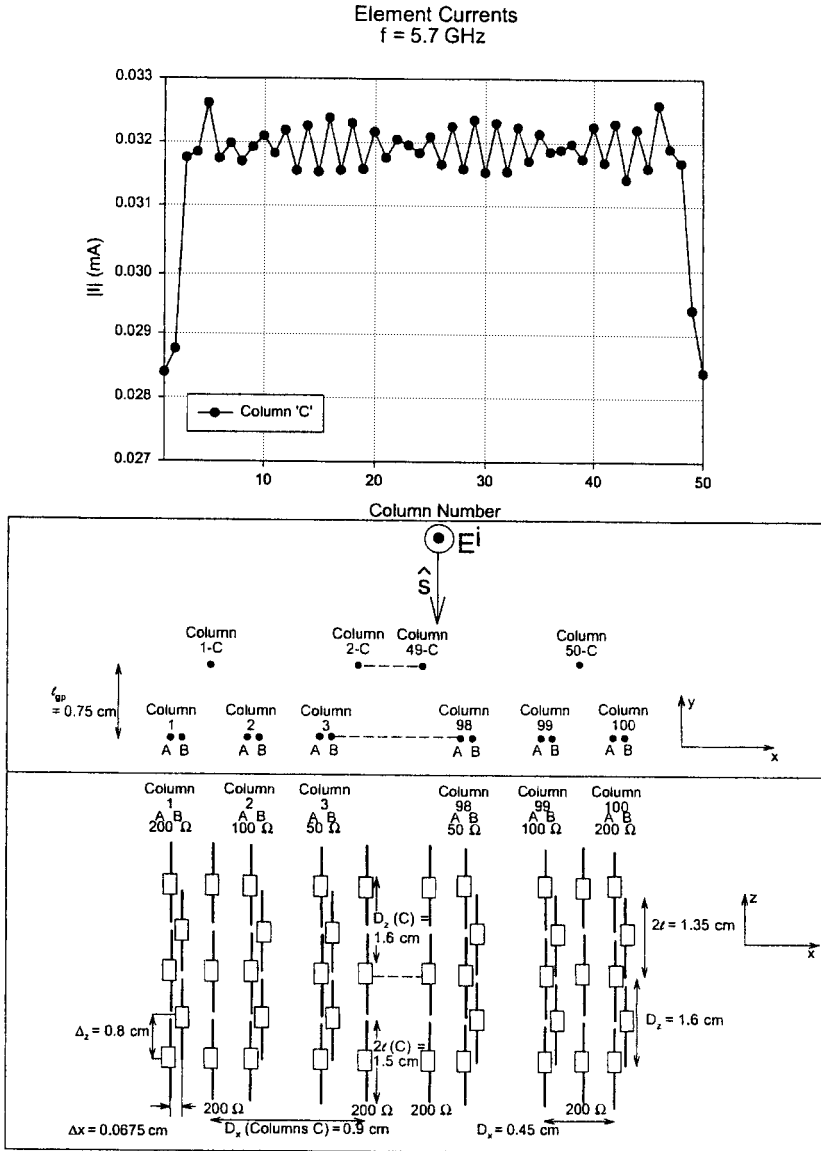


Fig. 5.14 The element currents of the active dipoles placed in front of the lightly loaded FSS array shown in Fig. 5.11. The same case as shown in Fig. 5.13, but this time all the active dipoles are loaded with 200 ohms. Frequency is 5.7 GHz.

a significant reduction of the current variations as expected. This experiment seems to show rather conclusively that it is the finite FSS groundplane that is the “troublemaker.”

However, this hypothesis must be tested at other frequencies.

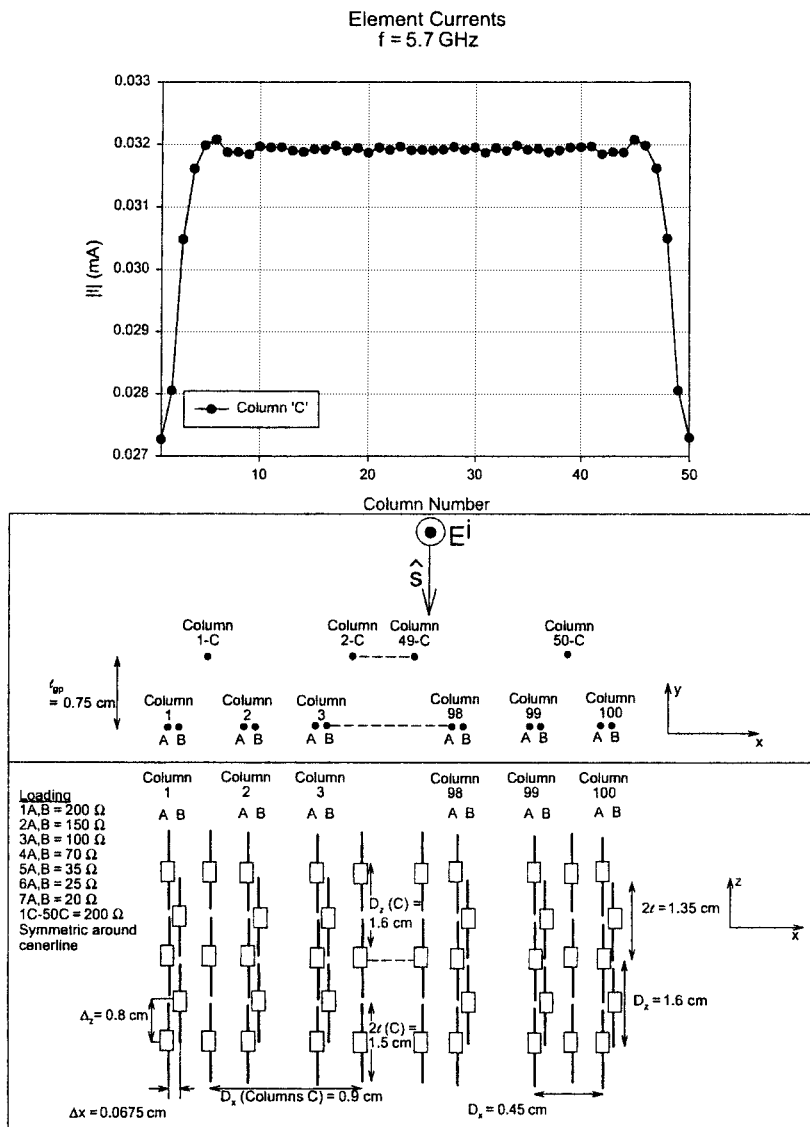


Fig. 5.15 The element currents of the active dipoles placed in front of the heavily loaded FSS array shown in Fig. 5.12. The active dipoles are loaded with 200 ohms. Frequency is 5.7 GHz.

5.7.2 Middle Test Frequency $f_M = 7.8 \text{ GHz}$

We next show the same three cases as in Figs. 5.13, 5.14, and 5.15, but at the middle frequency $f_M = 7.8 \text{ GHz}$. More specifically, Figs. 5.16 and 5.17 show the lightly loaded FSS groundplane with active load impedances $R_L = 100 \text{ ohms}$ and 200 ohms , respectively. This time we observe a significant reduction of the

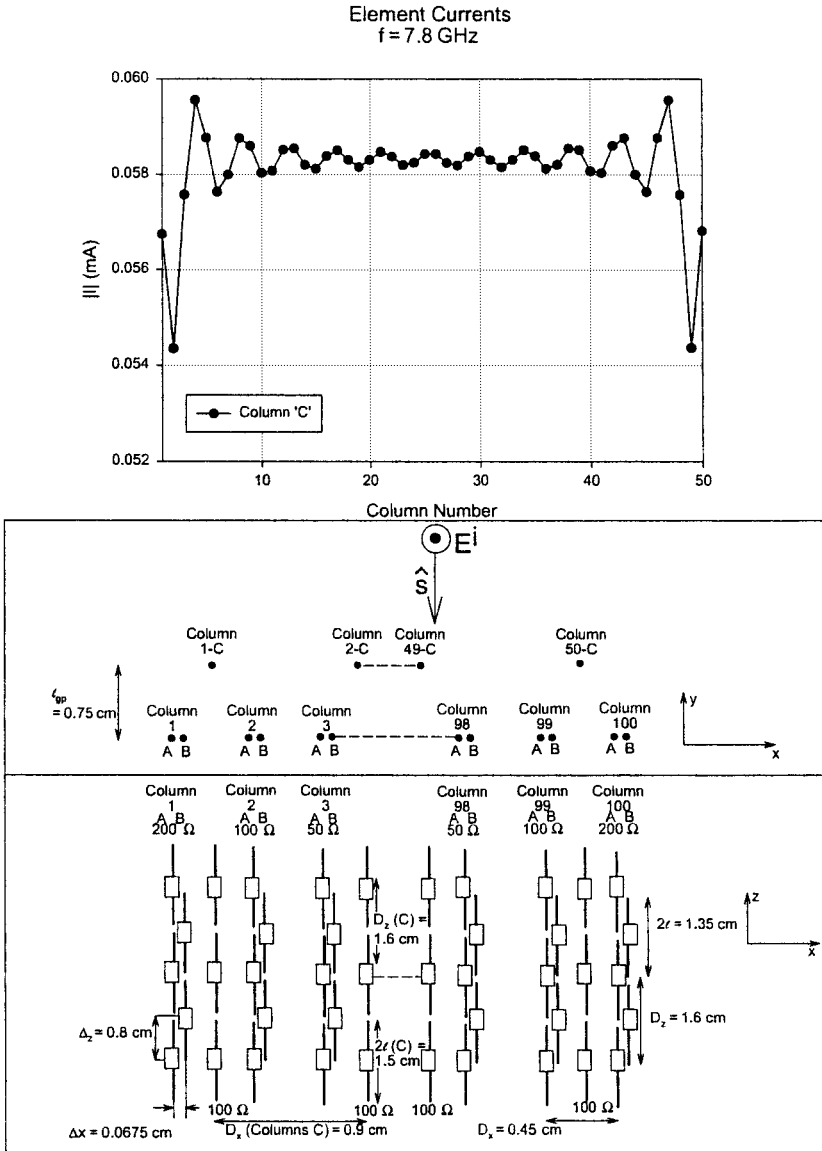


Fig. 5.16 The element currents of the active dipoles placed in front of the lightly loaded FSS array shown in Fig. 5.11. Active dipole loading is 100 ohms—that is, same case as Fig. 5.13 except frequency is equal to 7.8 GHz.

surface wave for 200 ohms as we would expect but did not observe in Figs. 5.13 and 5.14. Furthermore, we show the heavily loaded FSS case in Fig. 5.18 for load impedance $R_L = 200 \text{ ohms}$. Curiously, we do not observe much of an improvement except at the edge region.

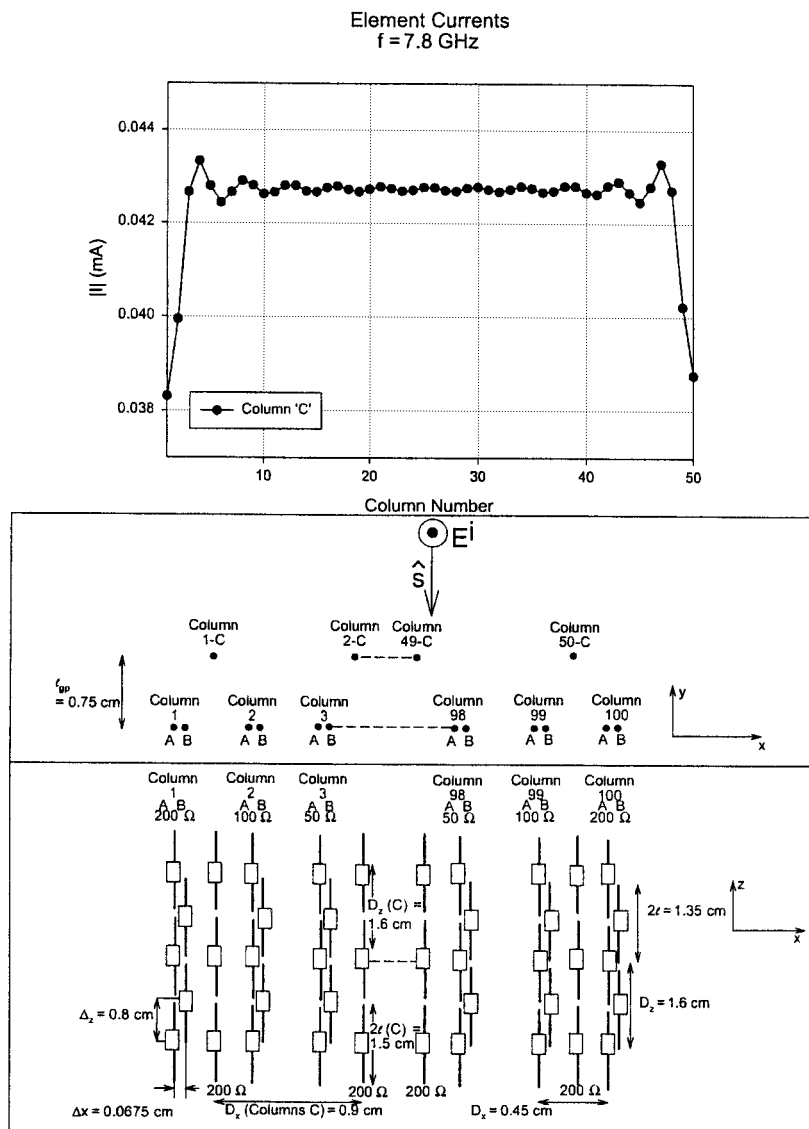


Fig. 5.17 The element currents of the active dipoles placed in front of the lightly loaded FSS array shown in Fig. 5.11. Active dipole loading is 200 ohms — that is, the same case as Fig. 5.14 except frequency is equal to 7.8 GHz.

5.7.3 High Test Frequency $f_H = 10 \text{ GHz}$

In Figs. 5.19, 5.20, and 5.21 we finally show the same three test cases as above but at the high test frequency $f_H = 10.0 \text{ GHz}$. Again, we show the lightly loaded cases in Figs. 5.19 and 5.20 with active loads being equal to $R_L = 100 \text{ ohms}$ and

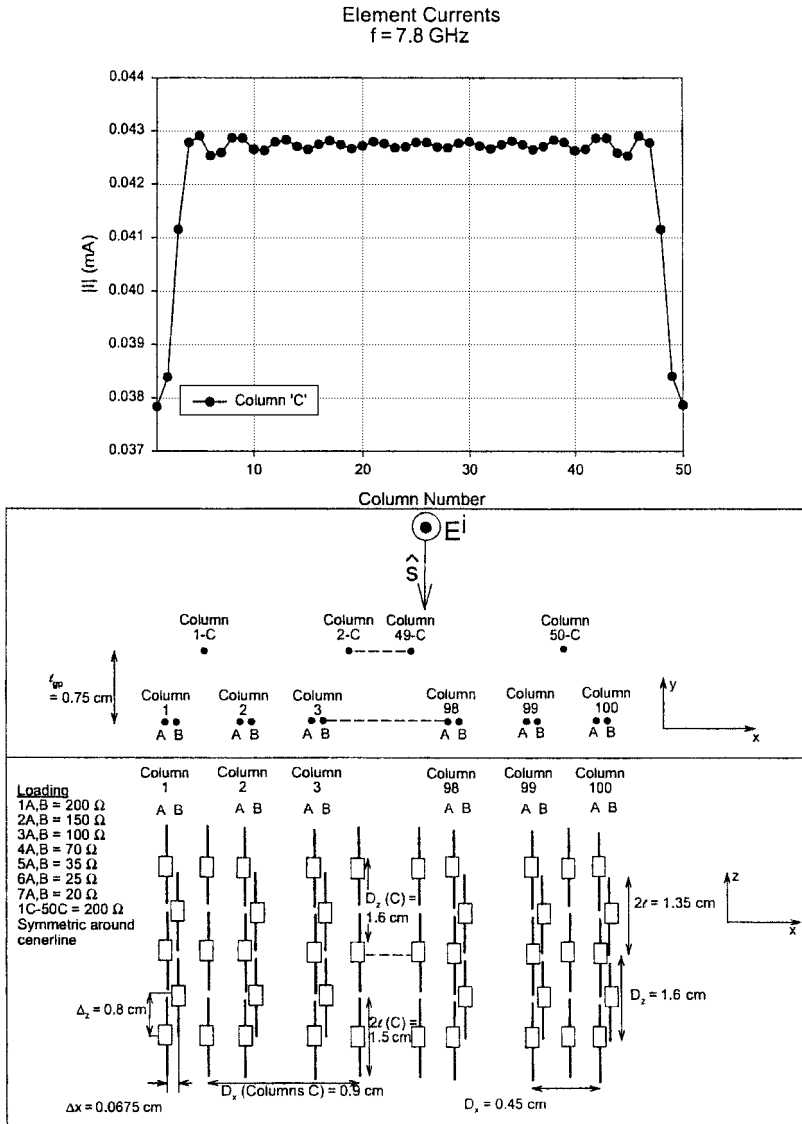


Fig. 5.18 The element currents of the active dipoles placed in front of the heavily loaded FSS array shown in Fig. 5.12. Active dipole loading is 200 ohms — that is, the same case as Fig. 5.15 except frequency is equal to 7.8 GHz.

200 ohms, respectively. As expected, we observe the most significant reduction of the surface waves when the active load resistance R_L is highest. Similarly, we show in Fig. 5.21 the heavily loaded FSS case when the load resistance R_L is 200 ohms. As we saw earlier above, the oscillations have not been reduced in the largest part of the array, only in the edge region.

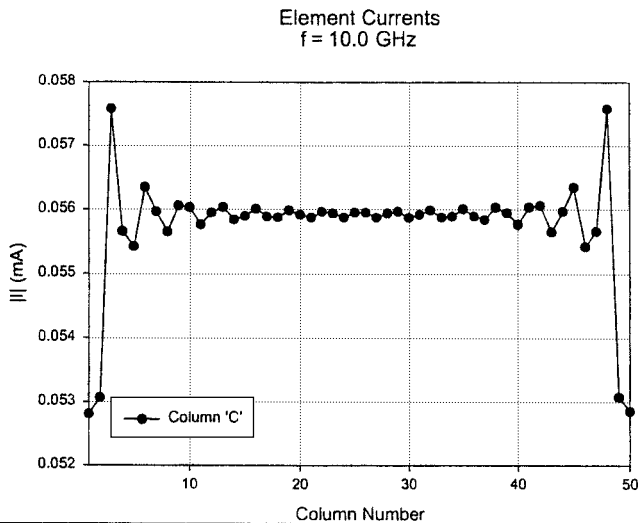


Fig. 5.19 Same case as shown in Figs. 5.13 and 5.16 except frequency is 10.0 GHz.

5.8 THE BACKSCATTERED FIELDS FROM THE TRIADS IN A LARGE ARRAY

The scattering diagrams at normal angle of incidence from an array consisting of 7 triads was shown in Fig. 5.8. All the active dipole elements in front of the FSS groundplane were loaded with identical resistors $R_L = 235$ ohms. We observed

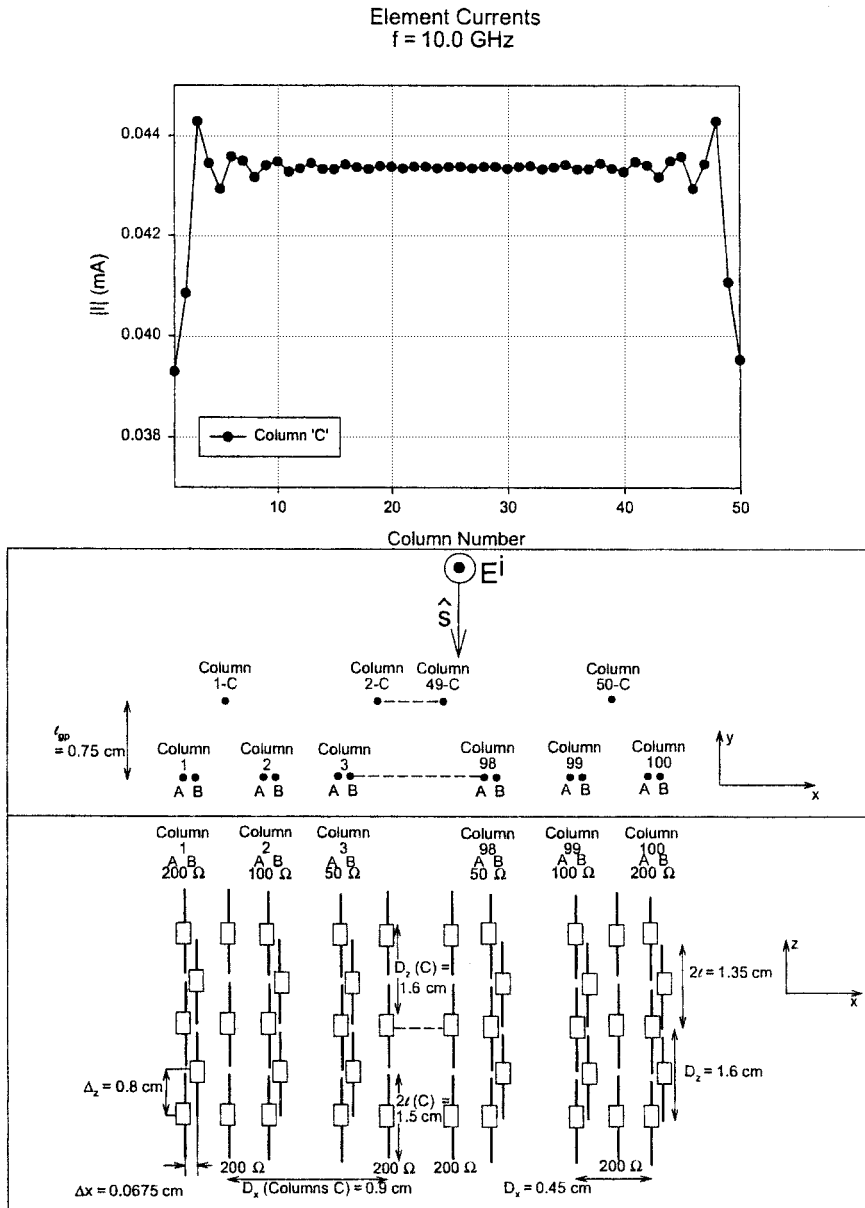


Fig. 5.20 Same case as shown in Figs. 5.14 and 5.17 except frequency is 10.0 GHz.

that the backscattered fields from the two edge triads differed significantly from the other triads. These, however, looked basically alike except for some minor deviations, often referred to as "jitter." This phenomenon appears to be typical for all finite periodic structures regardless of their extent. It has long been a source

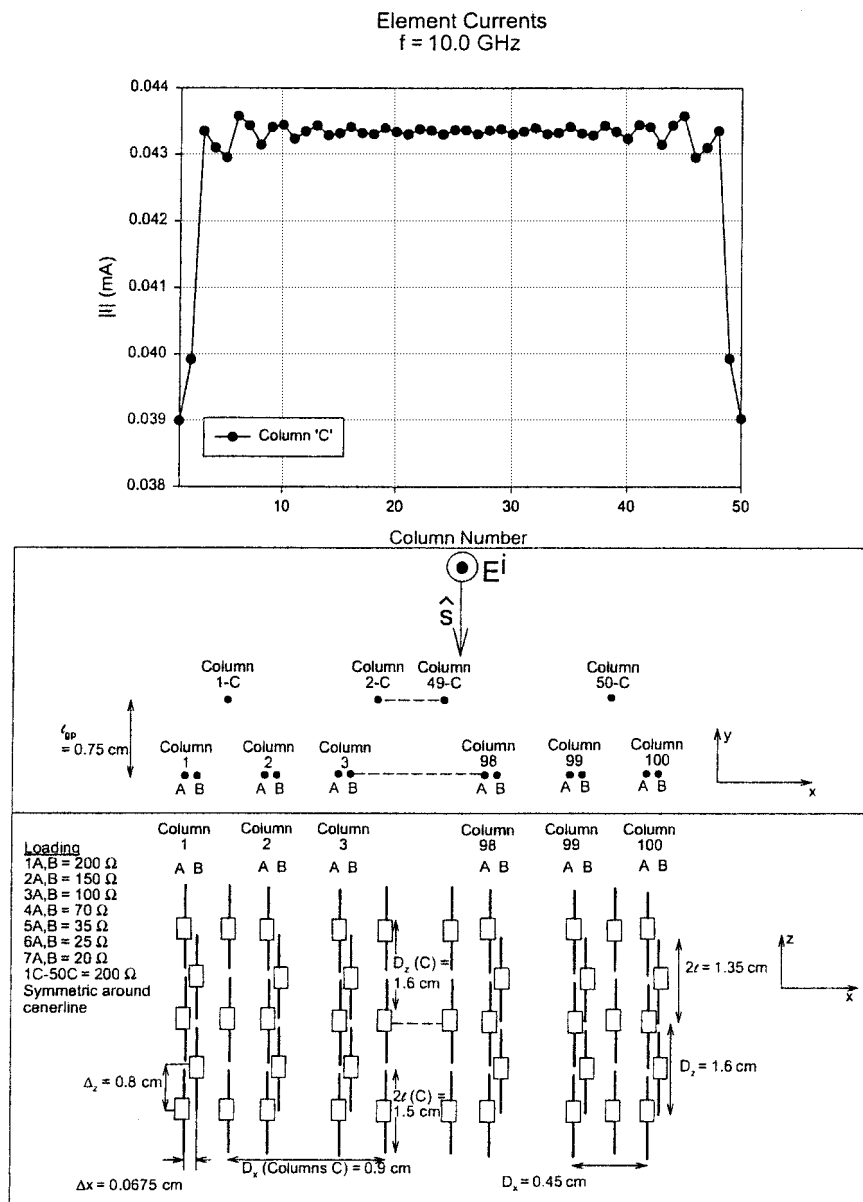


Fig. 5.21 Same case as shown in Figs. 5.15 and 5.18 except frequency is 10.0 GHz.

of irritation to designers of practical arrays since it makes precise matching of each triad with identical matching networks problematic. And as discussed in Chapter 2, superior matching is crucial if a low RCS at normal angle of incidence is desirable.

We postulated that this problem was associated with the presence of surface waves characteristic for finite structures. Thus, to reduce the jitter our aim should be to suppress these surface waves as much as possible.

We shall test this hypothesis on an array backed by the heavily loaded FSS groundplane shown in Figs. 5.15, 5.18, and 5.21. Thus, we show in Fig. 5.22 the scattering diagrams of 5 typical triads in an array of 50 triads. Each of the active elements are loaded with identical loads $R_L = 235$ ohms, and the distances between the active elements and the FSS groundplane are all equal to 0.68 cm. These values bring the backscatter at 10 GHz to the center of the scattering diagram; that is, we experience no backscatter at that frequency. However, we do at all other frequencies. To determine how much, we introduce the VSWR circles in the following way. We choose a VSWR equal to say m_1 and load the active elements with R_L/m_1 and $R_L m_1$ ohms, respectively (where as noted above $R_L = 235$ ohms). From the SPLAT program we then obtain the reflection coefficients for each of these values and plot them in the complex plane as shown in Fig. 5.22. These two reflection coefficients are located diametrically opposite in the complex plane and will consequently determine the circle corresponding to $\text{VSWR} = m_1$. For easy reference we also show in Fig. 5.22 the reflection coefficients (in decibels) corresponding to the respective VSWRs.

The triads depicted in Fig. 5.22 are nos. 10, 15, 23, 24, and 25—that is, typical specimens removed somewhat from the edges of the array. We observe that the reflected fields from all 5 triads in the frequency range 7–12 GHz are so close to each other that they practically melt together into single points. However, below 7.0 GHz this is certainly not the case. In fact, we show the scattering diagram for triad no. 23, 24, and 25 in Fig. 5.23 for the frequency range 5.0–7.0 GHz. Obviously, similarity between any of these 3 triads would be a complete coincidence. We strongly suspect that this phenomenon is directly related to the very strong presence of surface waves below 6.7 GHz in the FSS groundplane as observed clearly in the scattering diagram shown in Fig. 5.9. In other words, even heavy loading is not sufficient to completely eradicate the surface waves in the finite FSS groundplane at these lower frequencies.

However, we should by no means despair and conclude that operating a phased array with precision at the lower frequencies is a real problem. Recall that we are using the FSS groundplane merely to model a finite groundplane. When actually building a practical array, we would most likely use an ordinary piece of sheet metal of finite length and with holes so small that surface waves would be unlikely.

Which brings us right up to the very important question: What actually happens at the very edges of the finite array?

To answer that question we show in Fig. 5.24 the scattering diagram for the 5 triads closest to the edge, namely nos. 1, 2, 3, 4, and 5, respectively. Recall now that these triads, besides being at the edge, have the FSS portion right behind the active elements loaded heavily with resistors in order to attenuate potential surface waves (see Fig. 5.21). Thus, we would expect the scattering diagrams to differ somewhat from the “inner” triads shown in Fig. 5.22. Obviously, we

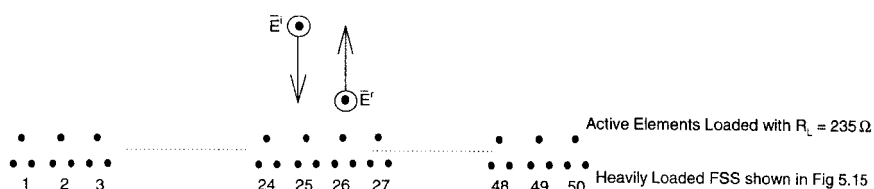
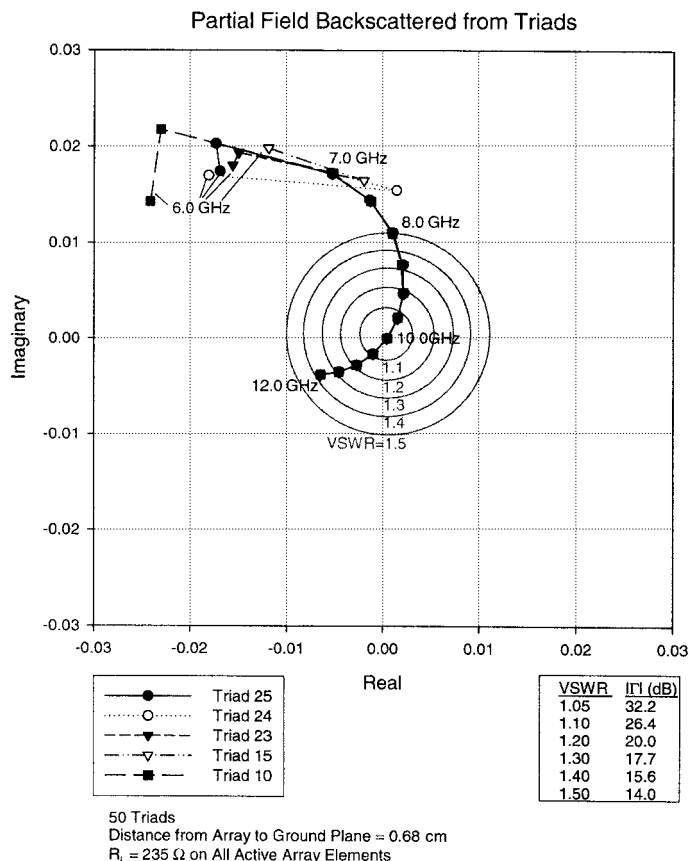


Fig. 5.22 The scattering diagrams for 5 typical triads as shown in an array of 50 triads. Note how the scattering from all 5 triads is virtually identical in the frequency range 7–12 GHz without any jitter as was the case in the untreated array in Fig. 5.8. Below 7 GHz we observe strong jitter as shown in detail in Fig. 5.23.

would like to reduce the backscatter from these outer triads as well. However, we are (or should be) prepared to use special R_L 's and design special matching networks for these triads that differ from the ones designed for the majority. This would just be considered part of the edge treatment we would expect to do for any finite structure.

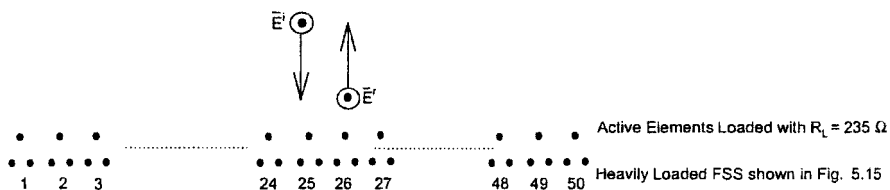
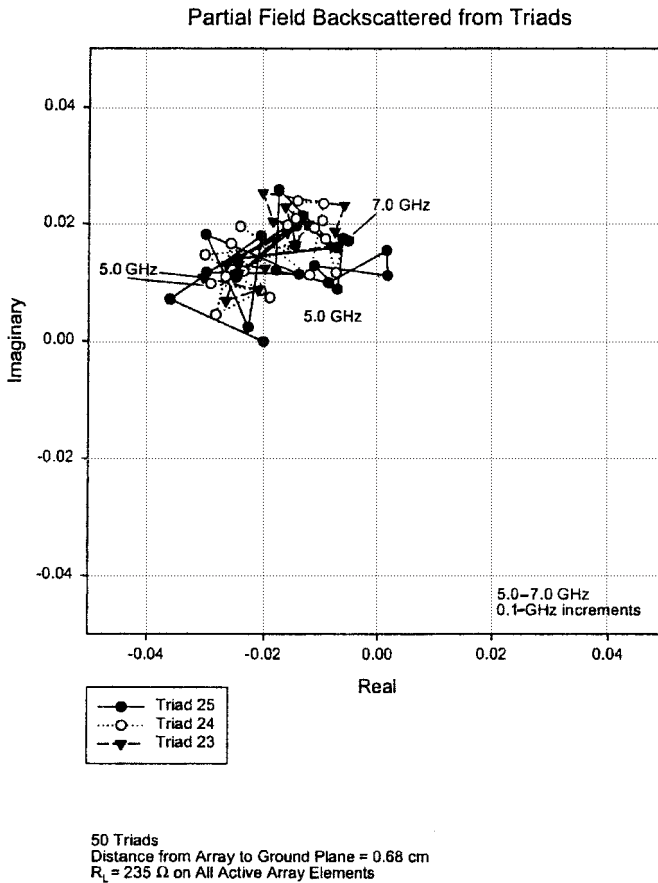


Fig. 5.23 The scattering diagram for the 3 triads in the center of the same array as in Fig. 5.22 but in the frequency range 5–7 GHz. The strong jitter is attributed to surface waves present in the FSS groundplane as shown in Fig. 5.9.

More specifically, we observe in Fig. 5.24 that triads 1 and 2 show the greatest deviation while triads 3, 4, and 5 more closely follow the majority shown in Fig. 5.22.

Furthermore, if triad 1 is loaded with 220 ohms rather than $R_L = 235$ ohms and the spacing between the active element and the FSS groundplane is decreased

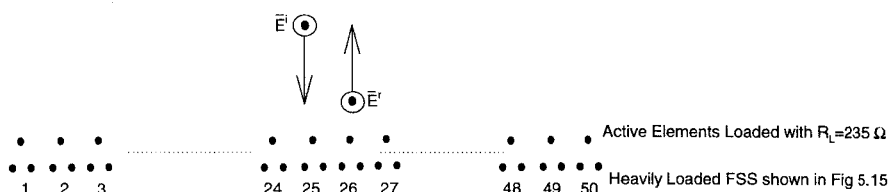
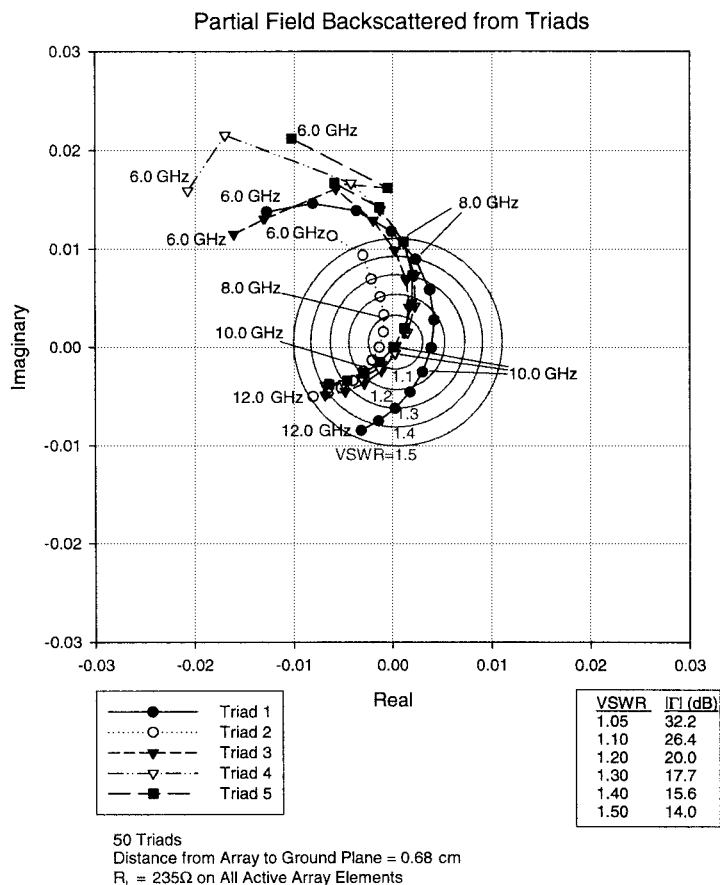


Fig. 5.24 The scattering diagram for the 5 triads closest to the edge of the same array as shown in Figs. 5.22 and 5.23. Note how the jitter is reduced as we move away from the edges for frequencies above 7 GHz but not below.

to 0.62 cm rather than 0.68 cm, we can as shown in Fig. 5.25 bring the 10-GHz point to the center while the other frequencies are now much closer to each other. Similarly, triad 2 can be brought closer to the center by increasing the load resistor to 280 ohms while we leave the spacing between dipole and FSS unchanged at 0.68 cm.

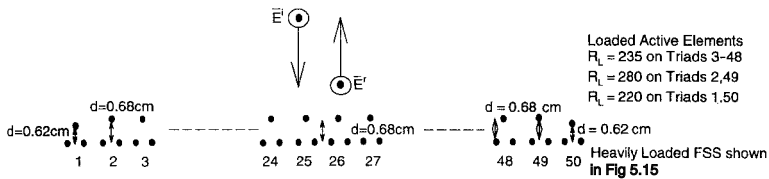
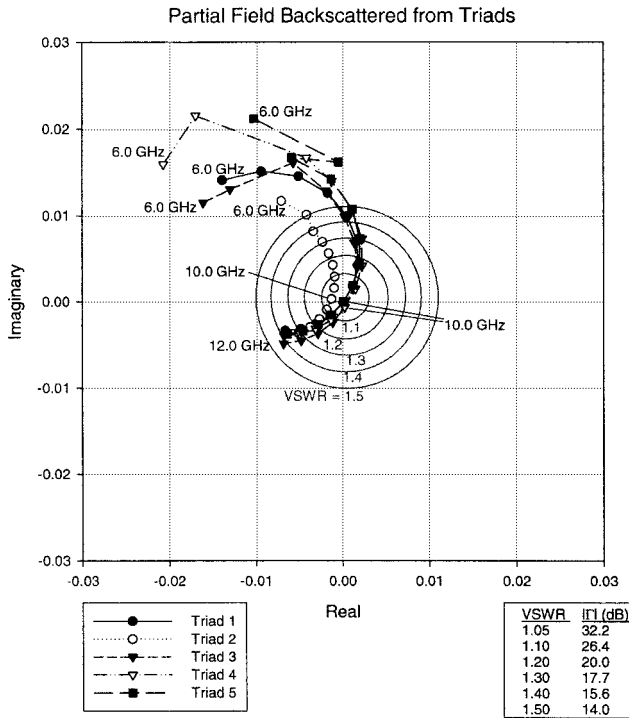


Fig. 5.25 By decreasing the load resistor to 220 ohms and reducing the spacing between dipole and the FSS to 0.62 cm, triad 1 can be moved close to the center. Similarly, triad 2 is loaded with 280 ohms.

5.9 ON THE BISTATIC SCATTERED FIELD FROM A LARGE ARRAY

In the previous section we determined the backscattered fields from the individual triads at normal angles of incidence. This information yields the RCS at boresight and is very important from a practical point of view.

It is, however, very instructive to observe how the field is scattered in all directions when the incident field is arriving from a fixed direction. Thus, we show at the top of Figs. 5.26 through 5.29 the bistatic scattered field at $f = 5.7$, 7.8, 10.0, and 12.0 GHz, respectively, for a typical triad (no. 25).

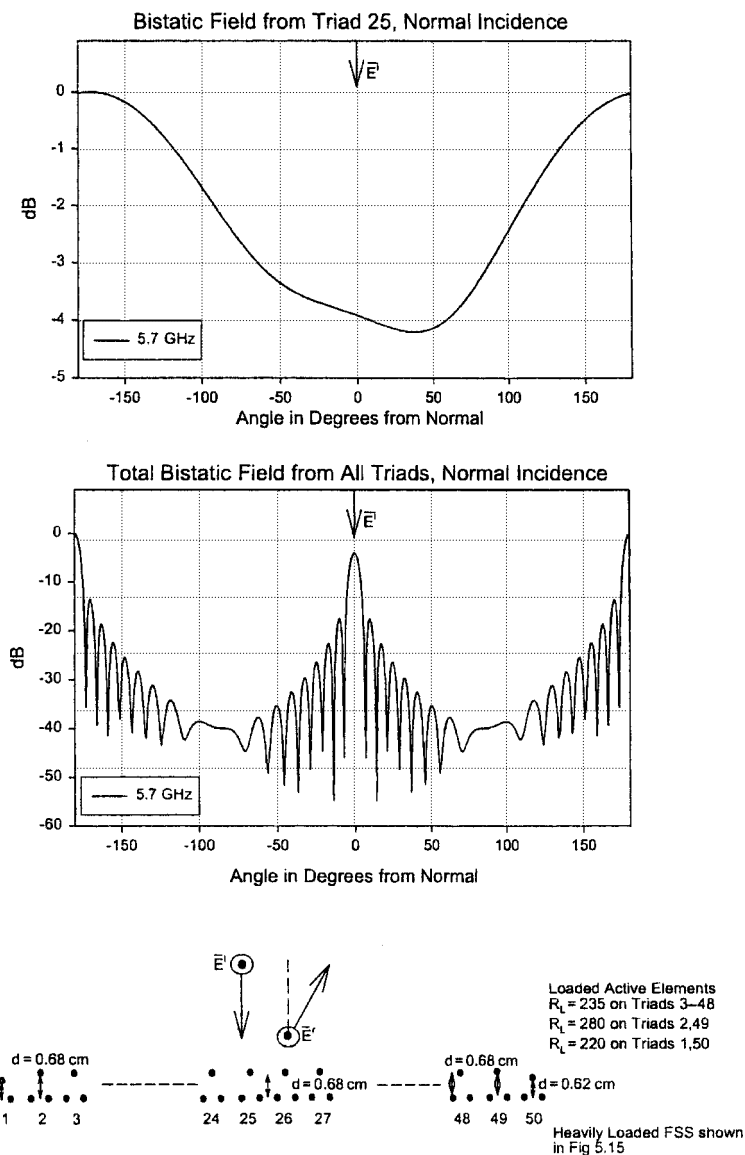


Fig. 5.26 Top: The bistatic scattered field from a single typical triad (no. 25). Bottom: The bistatic scattered field from the entire array of 50 triads. Normal angle of incidence. $f = 5.7$ GHz.

The direction of the incident wave is normal to the array. Note that the backscatter at the lowest frequency (5.7 GHz) is relatively high due to the inferior matching as observed in Figs. 5.22 and 5.23. However, at the design frequency 10 GHz the backscatter is very low as expected due to the superb matching as illustrated in Fig. 5.22.

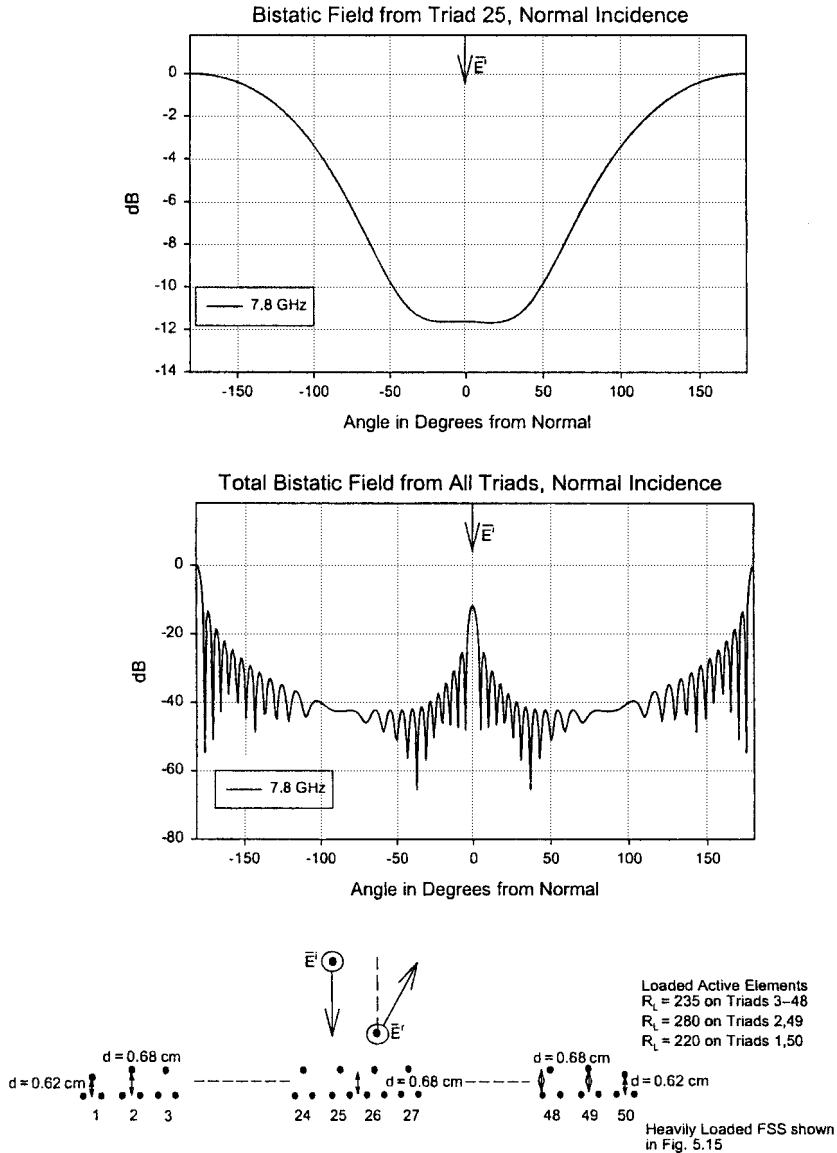


Fig. 5.27 Top: The bistatic scattered field from a single typical triad (no. 25). Bottom: The bistatic scattered field from the entire array of 50 triads. Normal angle of incidence. $f = 7.8 \text{ GHz}$.

If all the triads had identical bistatic scattering patterns, we could next obtain the total bistatic scattered field for the entire array by multiplication with the array factor. A typical example of this factor at 10 GHz is shown in Fig. 5.30. At other frequencies they look similar except for the beamwidth and are therefore not shown.

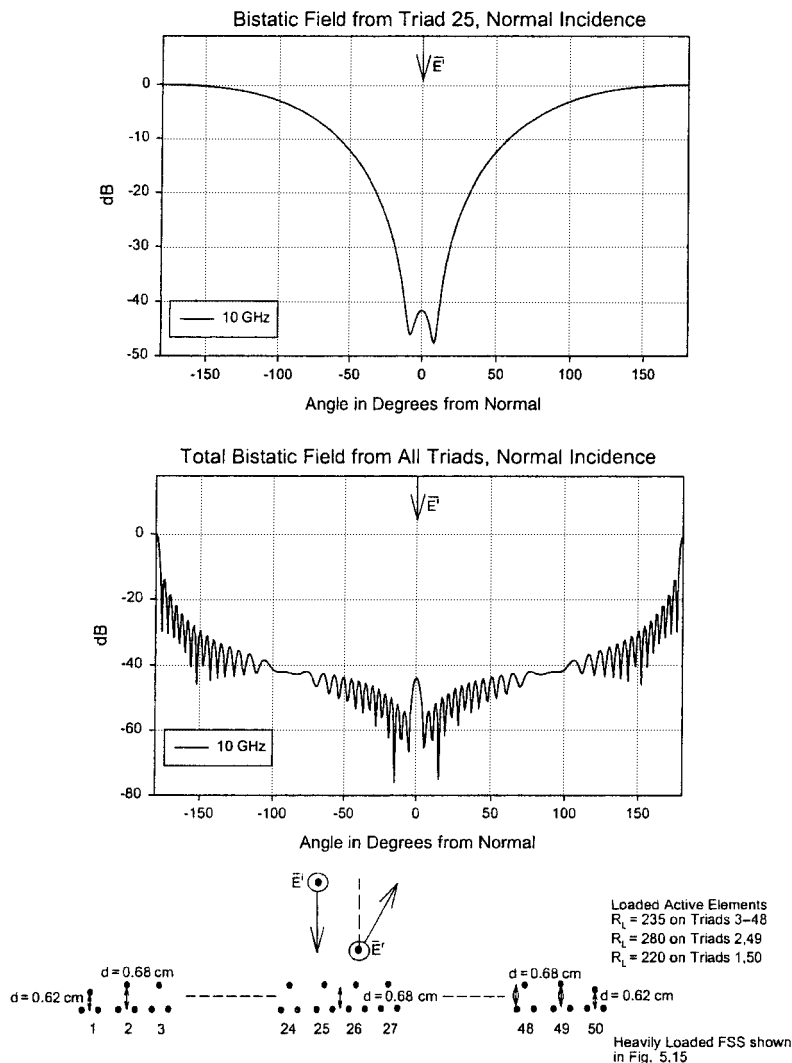


Fig. 5.28 Top: The bistatic scattered field from a single typical triad (no. 25). Bottom: The bistatic scattered field from the entire array of 50 triads. Normal angle of incidence. $f = 10.0 \text{ GHz}$.

However, the bistatic scattering patterns are as implied earlier not quite the same for the triads located close to the edges (see Figs. 5.22 and 5.25). Thus, simple array theory strictly speaking does not apply, although it is actually fairly accurate for large arrays with compensated edges as noted later.

We will therefore obtain the bistatic scattered field from the entire array by simply adding the bistatic scattered field from all the individual triads. Typical examples are shown at the bottom of Figs. 5.26 through 5.29 at the same frequencies as above, namely 5.7, 7.8, 10.0, and 12.0 GHz, respectively.

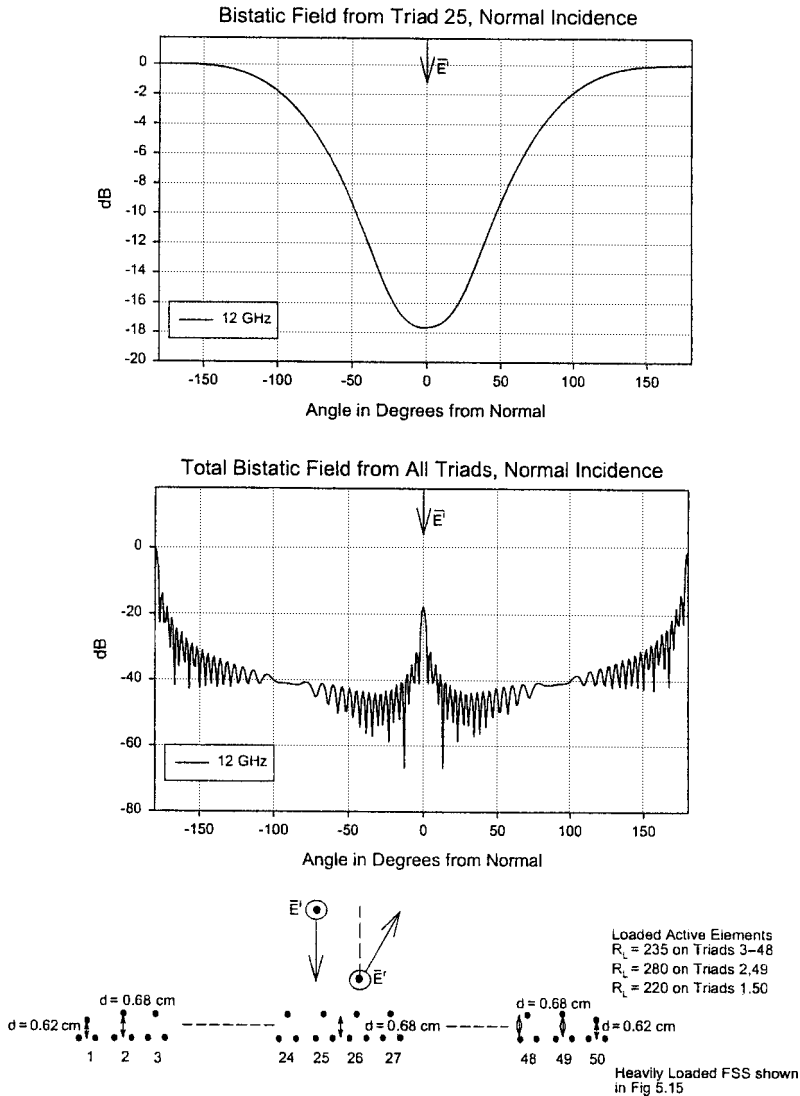


Fig. 5.29 Top: The bistatic scattered field from a single typical triad (no. 25). Bottom: The bistatic scattered field from the entire array of 50 triads. Normal angle of incidence. $f = 12.0$ GHz.

Again, as expected, we note a very high backscatter at the lowest frequency 5.7 GHz while we observe a very low backscatter at the design frequency 10 GHz.

Furthermore, we observe that the forward scattered field is at the same level as the backscatter of a perfectly conducting groundplane at the same size as the finite array. This is in agreement with our observations in Figs. 5.2 and 5.3.

Next, in Figs. 5.31 through 5.34 we show the bistatic scattered field for the same array as above but for angle of incidence equal to -30° . At the top we

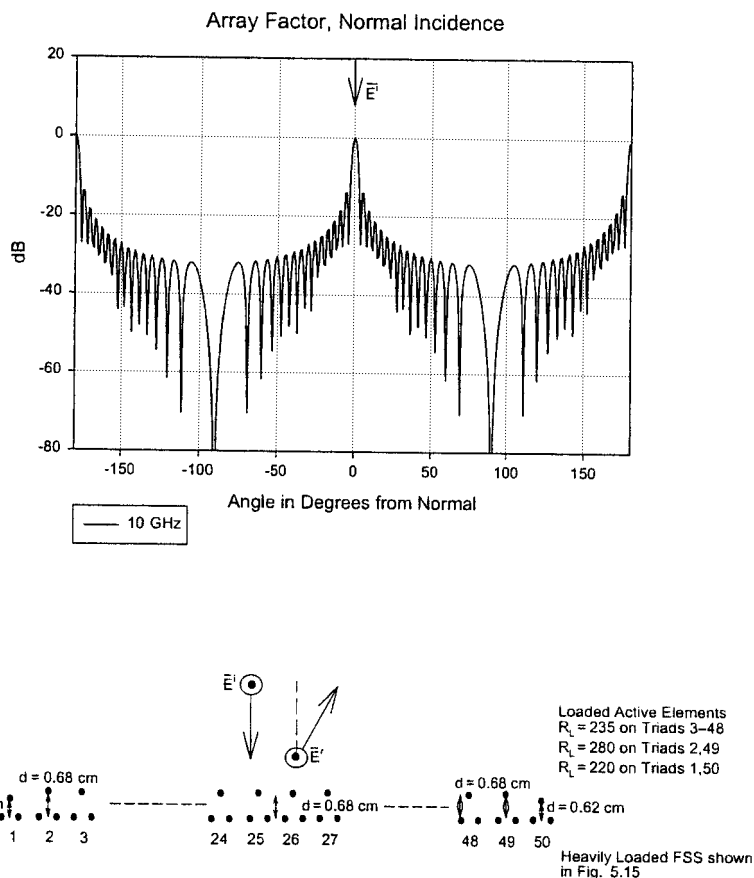


Fig. 5.30 The array factor for the entire array of 50 triads. Normal angle of incidence. $f = 10.0$ GHz. The factors at the other frequencies look quite similar except for beamwidths.

show the pattern for a single typical triad (no. 25), and at the bottom we show the bistatic scattered pattern for the entire array. Furthermore, in Fig. 5.35 we show a typical array factor for the entire array at $f = 10$ GHz and angle of incidence equal to -30° . The patterns at other frequencies look similar.

It is interesting to note that the backscatter level for oblique incidence is considerably lower than that for normal incidence due to the array factor.

Again we note that the level in the forward direction (150°) is 0 dB as it was for normal angle of incidence.

Finally we observe that the specular scattered field is quite closely equal to the normalized scattering value of a single triad only. Similarly, the backscattered field can be obtained as the sum of the backscatter from a triad and the array factor. This shows that for a large array with adjusted edge triads the total scattered field could have been obtained quite accurately by simple pattern multiplication of a typical triad and the array factor.

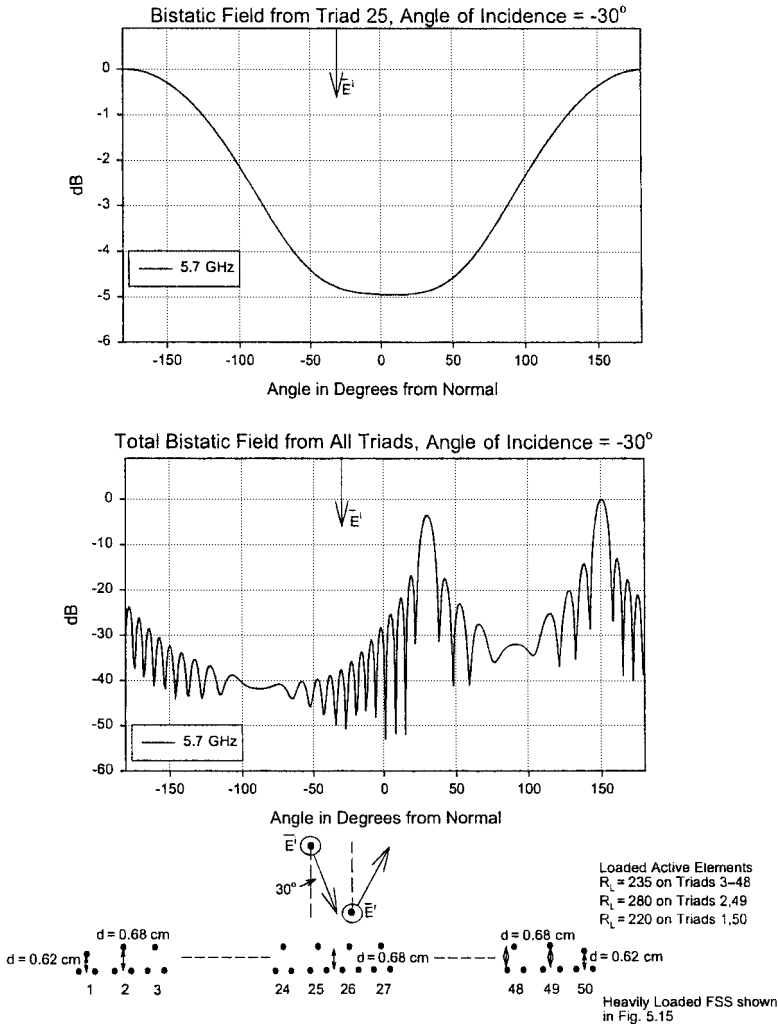


Fig. 5.31 Top: The bistatic scattered field from a single typical triad (no. 25). Bottom: The bistatic scattered field from the entire array of 50 triads. Angle of incidence is -30° . $f = 5.7$ GHz.

We may also state that the backscatter from a large array looks similar to that of an equal-sized groundplane from where we subtract the triad scattering.

To summarize: At broadside the backscattered field for a large array with compensated edges depends essentially on the scattering level of a typical triad. At oblique angle of incidence the total backscatter is given as the sum of the array factor and the triad scattering. Thus, even if the scattering pattern for a typical triad deteriorates somewhat for oblique incidence, the lower sidelobe level of the array factor can still be quite instrumental in producing a low backscatter for the entire array.

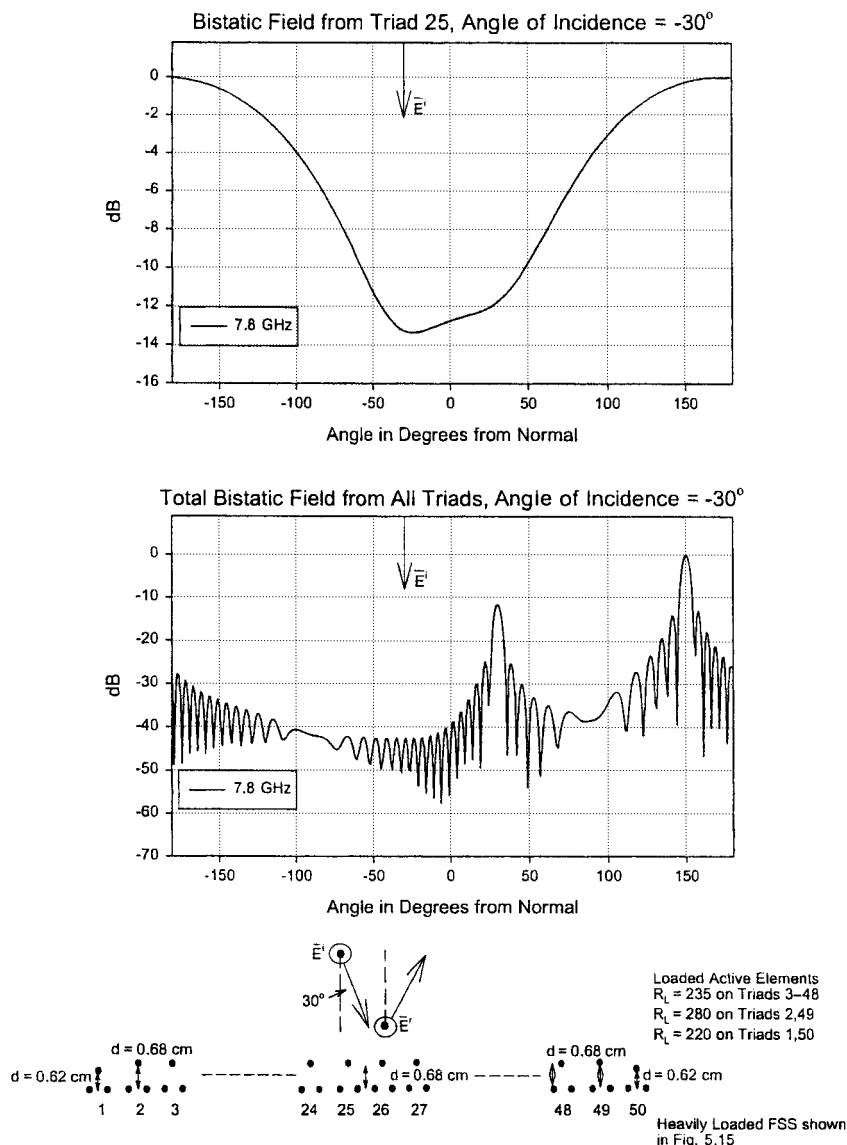


Fig. 5.32 Top: The bistatic scattered field from a single typical triad (no. 25). Bottom: The bistatic scattered field from the entire array of 50 triads. Angle of incidence: -30° . $f = 7.8$ GHz.

5.10 FURTHER REDUCTION: BROADBAND MATCHING

We demonstrated above that the backscatter is proportional to the reflection coefficient Γ . The array was matched at 10 GHz. At other frequencies Γ was larger, resulting in a stronger scattering from the individual triads.

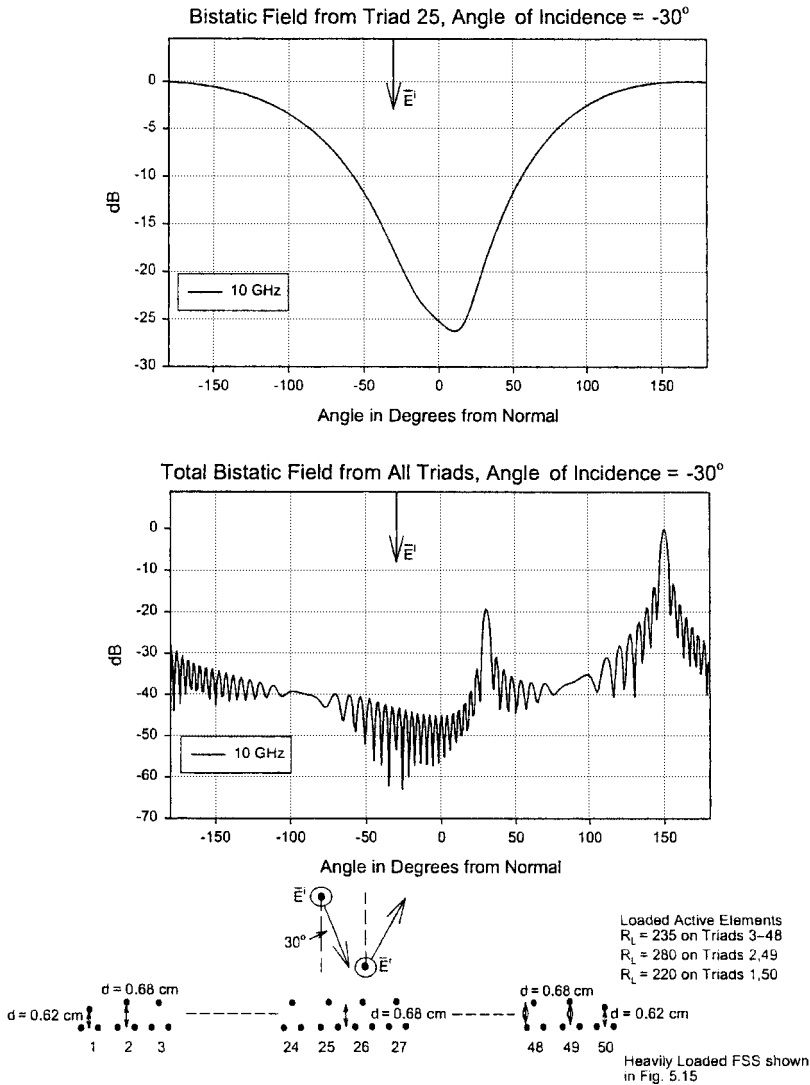


Fig. 5.33 Top: The bistatic scattered field from a single typical triad (no. 25). Bottom: The bistatic scattered field from the entire array of 50 triads. Angle of incidence is -30° . $f = 10.0$ GHz.

Thus, by better matching at all frequencies the backscatter could be further reduced. In other words we need to add a carefully designed matching network at the terminals of all the dipole elements. This network can be designed to be identical for all the typical triads since the array can be made free from jitter as seen earlier.

The active input impedance for triad 25 is shown in Fig. 5.36, bottom. It will be observed that the VSWR is ~ 1.02 at 10 GHz. In Fig. 5.36, top, we observe

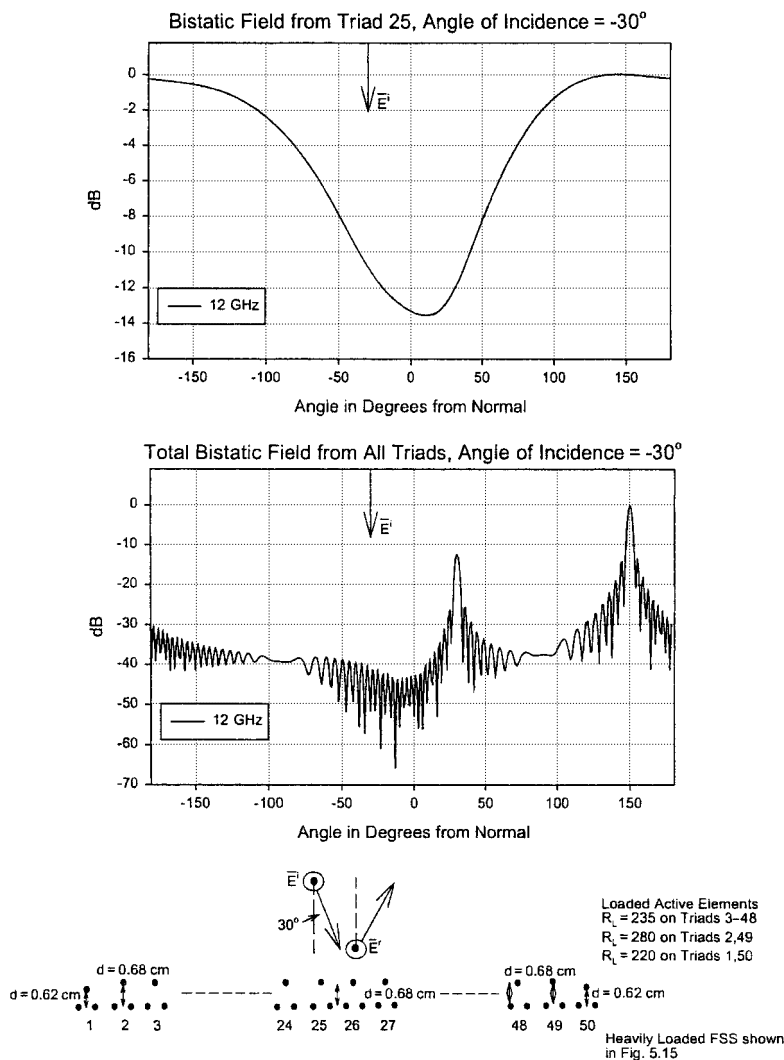


Fig. 5.34 Top: The bistatic scattered field from a single typical triad (no. 25). Bottom: The bistatic scattered field from the entire array of 50 triads. Angle of incidence is -30° . $f = 12.0$ GHz.

that the backscattered field at 10 GHz has a reflection coefficient corresponding to a standing wave ratio equal to ~ 1.01 . Considering that these calculations were obtained using only one mode, this agreement is quite satisfactory. See also Problem 5.3.

In Appendix B we review the principles for broadband matching, and in Chapter 6 we apply them to broadband arrays. In Problem B.3 you are invited to design a broadband matching network for the active input impedance shown in Fig. 5.36, bottom.

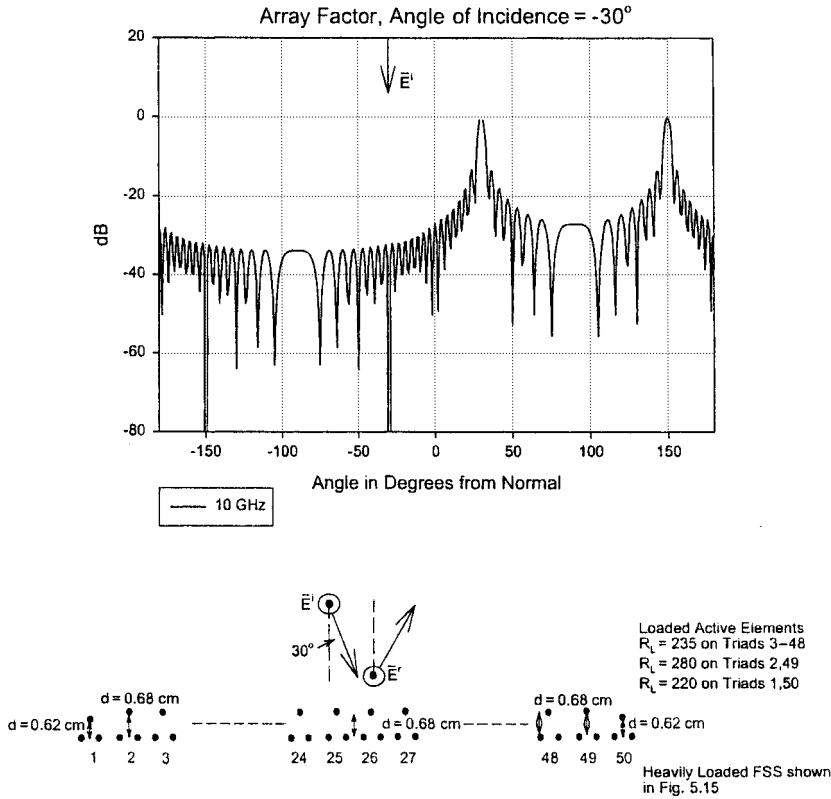


Fig. 5.35 The array factor for the entire array of 50 triads. Angle of incidence is -30° . $f = 10.0$ GHz. The factors at the other frequencies look quite similar except for beamwidth.

5.11 COMMON MISCONCEPTIONS

5.11.1 On Minimizing the Backscattering by Optimization

It is often suggested to minimize the backscattering from a finite array by a computerized optimization process. (Some people will go to great length to avoid any involvement with the physics of their problem. Some have no choice.)

While such an approach is feasible, it should be applied with great care. A very simple example will illustrate what can easily go wrong and undetected by operators whose intellectual capacity is limited to comparing numbers. In Fig. 5.37a we show the typical backscattered fields in vector form from each triad similar to the case in Fig. 5.8. Recall that the fields scattered from the two edge triads are quite different from the rest if all the triads are loaded with identical load resistors R_L . This was simply because the terminal impedances of the edge triads were in a different element environment, resulting in terminal impedances different from the rest.

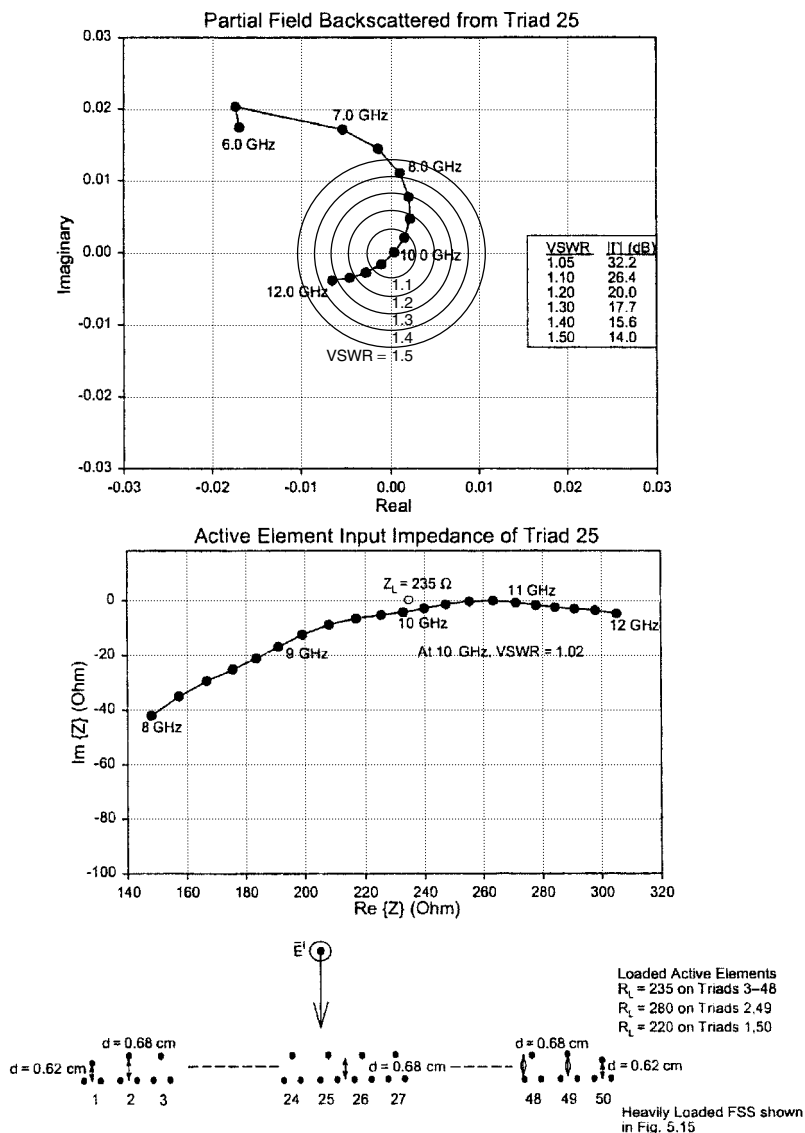


Fig. 5.36 Top: The backscattered field for triad 25 in an array of 50 triads. Bottom: The active input impedance for triad 25. Note that the incident field reflected from the outside very closely matches the magnitude of the reflection coefficient as seen at terminals from the inside. This shows that the residual scattering is ~ 0 .

When applying a computerized approach, we would most likely calculate the total backscattered field—that is, the sum of the triad fields. Typically a computer would then in all likelihood find the simple solution shown in Fig. 5.37b. By variation of the identical load impedances Z_L , the computer would produce a

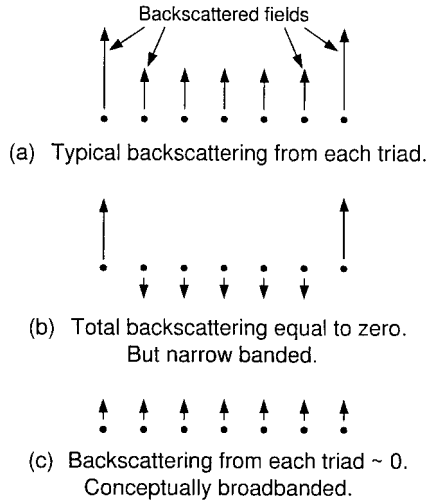


Fig. 5.37 Reducing the backscatter by an unguided optimization program typically produces inferior results. (a) Typical backscattering from the individual triads—in this case, strong at the edges and smaller elsewhere (see Fig. 5.8). (b) By varying the load impedances Z_L an optimization program calculating the total backscattered field will typically find a solution where the strong edge scattering simply cancels the weaker scattering from the triads elsewhere. However, the solution is not very broadbanded and is very sensitive to angle of incidence. (c) A much better solution is to reduce the backscatter from all triads and make them as close to zero as possible as discussed in Sections 5.4 and 5.9.

solution where the strong scattering from the two edge triads would simply cancel the sum of the much smaller fields from all the other triads. Thus, the total backscattered field could certainly easily be zero, but the result is obtained by simple cancellation which is in principle risky unless done with care. In this case the solution will be narrow-banded and highly sensitive to the angle of incidence.

A better solution is shown in Fig. 5.37c. Here the fields from *each* triad have been forced to be zero by the process discussed earlier in this chapter. It does not rely on simple cancellation and will consequently be more broadbanded and less sensitive to angle of incidence.

Oh yes, there is more to design than just run a computer, or worse yet, let the computer run itself without any interference from the human brain.

5.11.2 Can the RCS be Reduced by Treating the Dipole Tips?

In Problem 5.1 we address the fact that the backscatter does not always even appear to originate from the edges of a reflecting surface.

A related misconception is the notion that the backscatter from an array is originating at the very tips of the dipoles. Thus, it is often reasoned that if the tips could somehow be treated in some magic fashion, the backscatter from the array could possibly be reduced.

A common attack along this idea is to simply place resistors at the very tips of the dipoles. However, since there is basically no conducting current in this area, anything placed here will be practically ineffective (except possibly a small change in resonance frequency due to a slight increase in the end capacitance of the dipoles).

In order for a resistive loading to be effective, it must be inserted at a place where there is a current. And there must be a groundplane.

And that happens to be precisely what we do when we place load resistors in the middle of the dipoles as amply demonstrated in this chapter.

Asymptotic methods and way of thinking work fine at high frequencies when applied correctly. But they should not blindly and religiously be carried into the *resonance* region. They may “work” in some cases, but often merely due to luck and by coincidence.

5.12 CONCLUSION

In Chapter 2 we introduced the reader to the fundamentals of the RCS of antennas in general. We demonstrated how the scattering from any antenna could be decomposed into two components, namely the antenna mode component proportional to the mismatch at the terminals of the elements and another component today usually referred to as the residual scattering component. (For further discussion on this subject see Section 2.2.)

Array antennas with groundplane were of particular interest since they turned out to have no residual scattering component, at least potentially. Thus, to obtain zero backscatter we merely would have to match each element perfectly.

While this sounds conceptually simple, there are practical problems to be solved.

First, the triads in the edge area will have terminal impedances significantly different from all the other triads. This is simply related to the fact that the elements at the edge are in a different environment as far as mutual impedances are concerned. Thus, if the inside triads are conjugate-matched and the edge triads loaded with the same impedances, mismatch will occur, causing a significant amount of scattering from the edge columns.

The second problem is actually more complex. When considering an infinite array, the terminal impedance will be the same from element to element in accordance with Floquet’s Theorem. However, when the array is finite, it is well known that the terminal impedance will differ from element to element in an oscillating way around the infinite array value (sometimes denoted as jitter). We postulated that this phenomenon was related to the presence of surface waves of the same type as encountered in Chapter 4. However, there is a significant difference in amplitude of these surface waves in the passive and active cases. This is due to the fact that the elements in the former case in general are loaded with pure reactances (if any), while the elements in the latter case are (or should be) connected to individual amplifiers or generators containing substantial resistive components (as encountered when conjugate matched).

These resistive components cause significant attenuation of potential surface waves along the structure. In fact, they will in general be so weak that the surface wave radiation from active arrays can be ignored in contrast to the FSS case discussed in Chapter 4. However, they may be strong enough to produce jitter of the terminal impedance.

Furthermore, one more component will complicate the situation, namely the groundplane that in our investigation is modeled in the form of a finite FSS surface. Potential surface waves on such a structure cannot be attenuated by simple resistive loading all the elements across the entire surface since such an approach would lead to excessive reflection loss as a groundplane. Thus, possible surface waves must in this case be controlled in the same manner as was done in Chapter 4, namely by resistively loading just a few columns in the edge areas.

By monitoring the column currents as a function of the loading in the edge areas, we were able to obtain a significant reduction of the surface waves in the finite FSS groundplane. By further loading the active element of each triad everywhere with load resistors R_L equal to $R_A = 235$ ohms as discussed already above, we were able to eliminate all visible jitter in all triads at least between nos. 10 and 40 in the frequency range 7–12 GHz (and most likely significantly beyond 12 GHz). However, below 7 GHz we did indeed experience a significant amount of jitter. We attributed this almost entirely to the finite FSS groundplane. Thus, we could avoid this problem by lowering the resonant frequency of the FSS groundplane. Or better yet, simply realize that when building an actual array, we would most likely use a finite solid groundplane rather than a finite FSS groundplane. Thus, there would be no surface wave attached to the finite groundplane per se. At any rate, if we can avoid jitter when using a “nervous” FSS groundplane, we should be able to handle quite a few options.

Some deviations in scattering was observed from a few triads in the edge area. However, by adjusting the load resistors as well as the spacing to the groundplane, it was possible to reduce the scattering at the center frequency 10 GHz to zero while the scattering at the rest of the frequencies were sufficiently close to the triads in the rest of the array.

Let us summarize: It is possible by careful design to obtain large arrays of dipoles with a groundplane that are virtually free from jitter over a significant frequency range.

And once demonstrated for dipoles, it is generally recognized that it should be possible for other types of phased arrays as well.

The designs presented in this chapter have not been optimized for greater bandwidth essentially for two reasons:

1. We wanted primarily to demonstrate that jitter-free arrays simply exist.
2. Most practical arrays will be blended into an airframe or ship structure and should consequently be designed with this in mind.

Super-broadbanded infinite arrays will be discussed in Chapter 6.

PROBLEMS

- 5.1** It has often been noted that the scattering from a flat plate appears to emanate from the edges. This observation is certainly true when considering a perfectly conducting plate and to some extent a dielectric slab.

However, reflecting flat plates in general can be considerably more complicated than just that, and great care should be observed.

Discuss how it is possible to design flat finite reflectors that:

- Scatter strongly from the edges and absorb everywhere else.
- Absorb at the edges and reflect strongly everywhere else.
- Absorb at the edges and everywhere else.

So much for simplistic concepts.

- 5.2** Based on the curves given in Figs. 5.26 through 5.35, estimate the backscattered field for the entire array for angle of incidence equal to -15° .
- 5.3** From Fig. 5.36, bottom, calculate the VSWR at $f = 8, 9, 10, 11$, and 12 GHz. Compare these values with the standing wave ratio shown in Fig. 5.36, top, for the field reflected from the front of the array in the backscatter direction. Considering that only a single current mode is used, you will find the agreement quite satisfactory.